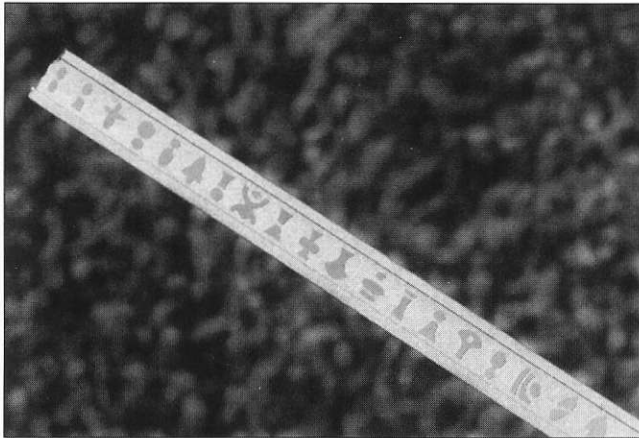




PROBING THE ROSWELL THIN-STRUT DEBRIS

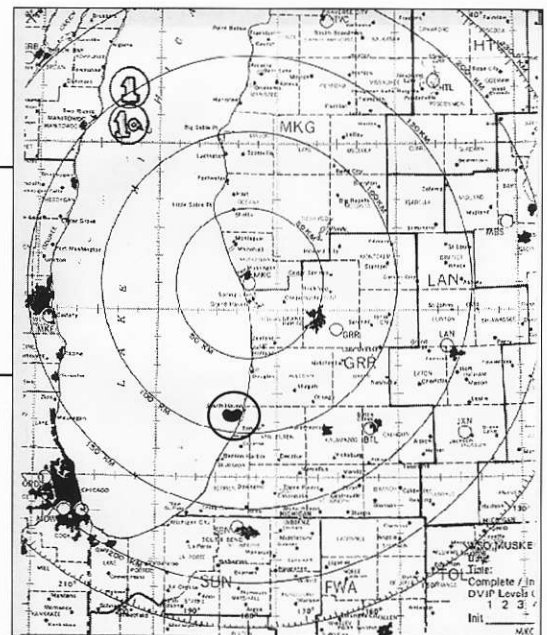
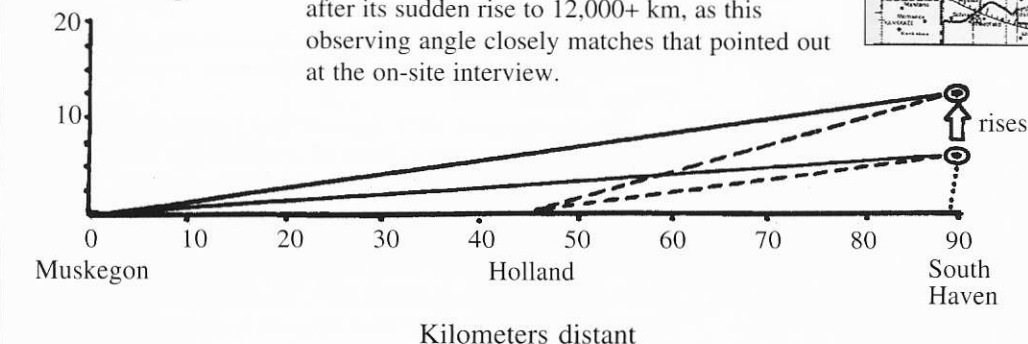
Milled-aluminum I-beam replica designed by Miller Johnson based on Jesse A. Marcel Jr.'s recollection. Photograph by Miller Johnson.

THE HOLLAND, MICHIGAN, RADAR-VISUAL CASE, 1994

- The line of sight of the radar
- - - The line of sight of the officer*
- The possible line of sight of the retired USAF colonel

Kilometers high

* The officer probably observed the phenomenon after its sudden rise to 12,000+ km, as this observing angle closely matches that pointed out at the on-site interview.



INTERNATIONAL UFO REPORTER

Editors:

Jerome Clark
George M. Eberhart
Mark Rodeghier

Contributing Editors:

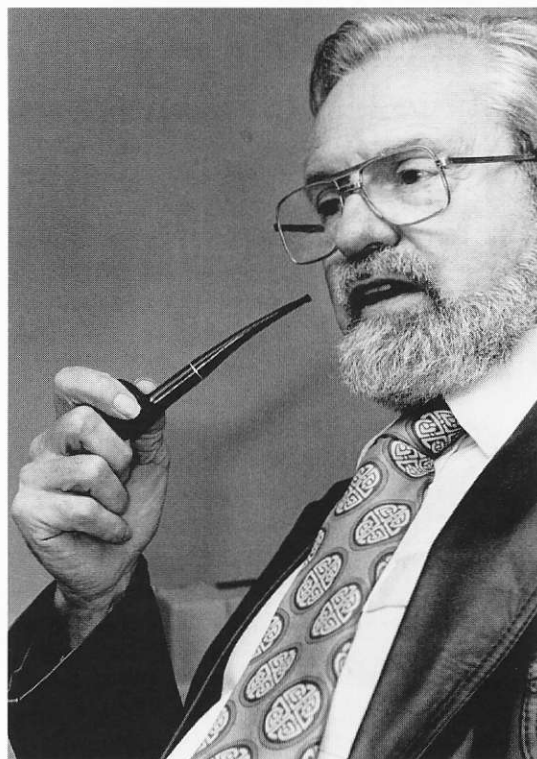
Bill Chalker
Richard F. Haines
Richard Hall
Kevin D. Randle
Jenny Randles
Chris Rutkowski
Jennie Zeidman

Web site:

www.cufos.org

E-mail:

Infocenter@cufos.org



J. Allen Hynek
1910-1986

THE HOLLAND, MICHIGAN, RADAR-VISUAL CASE	by Michael D. Swords	3
PROBING THE ROSWELL THIN-STRUT DEBRIS	by Robert A. Galganski	8
ASTRONOMERS AND UFOs	by Henry Cordova, Frank John Reid, and George M. Eberhart	13
FIRE ON THE MOUNTAIN: THE BERWYN MOUNTAIN INCIDENT	by Andy Roberts	16
THE MYSTERY OF HOWDEN MOORS: PART TWO	by David Clarke and Martin Jeffrey	24
LETTERS		29
OF INTEREST TO CUFOS ASSOCIATES		31

International UFO Reporter (ISSN 0720-174X) is published quarterly by the J. Allen Hynek Center for UFO Studies, 2457 West Peterson Avenue, Chicago, Illinois 60659. All rights reserved. Reproduction without permission is strictly prohibited. Copyright © 1999 by the J. Allen Hynek Center for UFO Studies. Third-class postage paid at Chicago, Illinois.

Advertisements accepted for publication in this magazine do not necessarily reflect the viewpoints of the J. Allen Hynek Center for UFO Studies.

Address all article submissions, letters to the editor, and other editorial correspondence to *International UFO Reporter*, Center for UFO Studies, 2457 West Peterson Avenue, Chicago,

Illinois 60659. Address all subscription correspondence to *International UFO Reporter*, 2457 West Peterson Avenue, Chicago, Illinois 60659.

The *International UFO Reporter* is a benefit publication mailed to Associates of the Center for a contribution of \$25.00 or more. Foreign Associates add \$5.00 for delivery. All amounts in U.S. funds. Other publications also available for contributors of larger amounts. For details, write to the J. Allen Hynek Center for UFO Studies, 2457 West Peterson Avenue, Chicago, Illinois 60659, USA. **Postmaster:** Send Form 3579 to CUFOS, 2457 West Peterson Avenue, Chicago, Illinois 60659.

THE HOLLAND, MICHIGAN, RADAR-VISUAL CASE, 1994

BY MICHAEL D. SWORDS

One of the best-witnessed radar-visual cases in UFO history occurred more than five years ago in western Michigan, but it has been poorly documented and described in the UFO literature. I attempt to remedy that situation with this article, describing the findings of our CUFOS investigation as well as the work of others. The CUFOS investigation was incomplete, for reasons that will become apparent, but we feel that this case deserves wider attention because of its potential importance.

THE STORY IN BRIEF

On the evening of March 8, 1994, a clear night, citizens in and around Holland, Michigan (on the shore of Lake Michigan), began observing lights in the sky about 9:15 p.m. To some, the lights appeared to be attached to one another, or at least coordinated in their movement. For at least one family, the Graveses, the experience was more of a close encounter with a disk with lights turning clockwise on its underside.

Several people phoned 911 to report the sightings, which led to the dispatch of Holland police officer Jeff Velthouse to investigate. Velthouse saw the lights himself, in part through binoculars supplied by the Graves family. After a short while, the Allegan County sheriff's office called the Muskegon National Weather Service (NWS) radar station about 30 miles to the north and asked them to see if they had any radar returns corresponding to the visual sightings. They did indeed, and the radar operator gave a live report, which was recorded by the police, of the behavior of the radar returns he was tracking.

The visual sightings lasted until about 11:00 p.m., while the unusual images were seen on the weather radar for about 20 to 30 minutes. We know that at least three dozen people saw the lights, although there is no doubt many others did but never came to the attention of UFO investigators.

These sightings received almost immediate national

media attention, in all types of outlets. For example, the Graves family appeared on the CBS program *48 Hours*. This is undoubtedly because the 911 phone calls were taped and released by the police, along with the recording of the frankly skeptical (then delightfully excited) NWS radar operator who wowed himself (and us) in real time. Many other sorts of witnesses ultimately emerged in other parts of western Michigan in the following days, as Michigan, both the state and lake, seemed to have lit up with UFOs that evening.

AFTERMATH

The presence of such willing witnesses as the Graves family, as well as the tape of the radarman, drew media like piranhas to a kill. A major circus immediately developed that greatly hampered investigations. Some people were harassed to the point of informing the local sheriff's office that in no way were their names to be given out, or any interactions with investigators facilitated. The Graves family seemed to enjoy their 15 minutes of fame, but many others wanted no part of it.

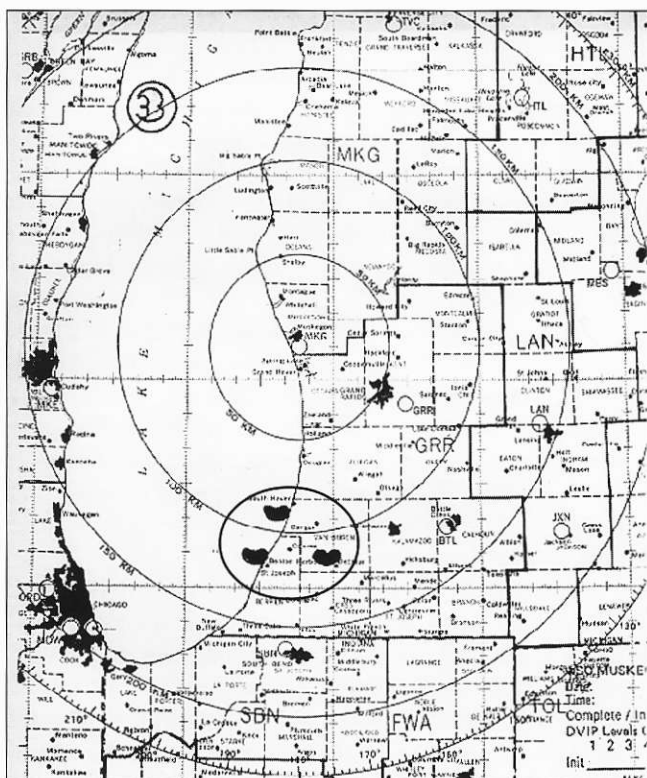
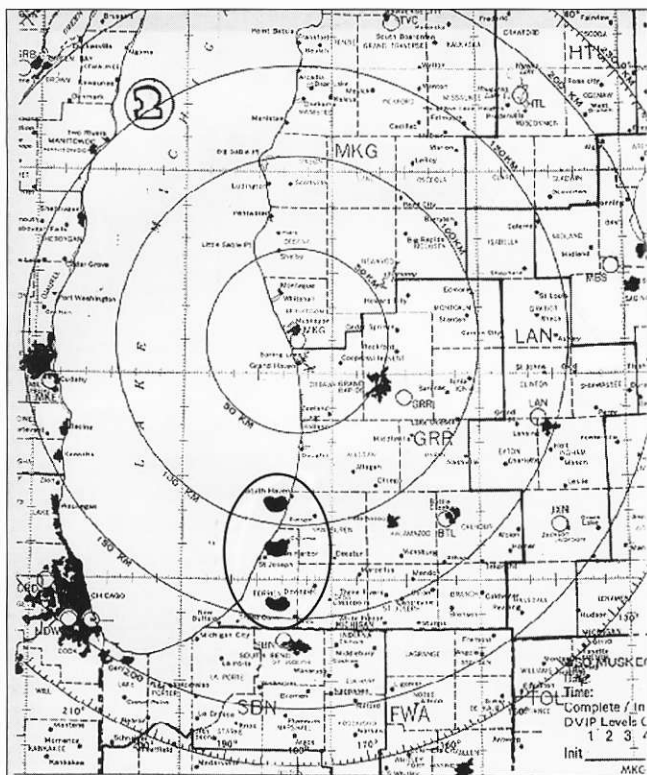
Investigators from the Michigan chapter of MUFON did an investigation, and *some* of the results of that work appeared in the *MUFON UFO Journal* in a September 1994 article, authored by then-Michigan MUFON President Shirley Coyne. A competent journalist from the *Muskegon Chronicle*, Mike Walsh, was also doing what he could, and he published short articles, including one for the May/June 1994 issue of *UFO Magazine*.

However, an investigative setback occurred when a witness who was a commercial pilot had his name mentioned on national television, thus getting him into trouble with his airline and closing off the opportunity to get details on an independent air witness to the events. Still, with many willing witnesses, the local police officer, and the audiotape, it was a fine media event and a splashy UFO story.

CUFOS ON THE CASE

CUFOS Scientific Director Mark Rodeghier learned of the events soon after the sightings occurred, and he called a friend of mine in the Kalamazoo area, an old NICAP

Michael D. Swords is professor of natural sciences at Western Michigan University, Kalamazoo, and former editor of the Journal of UFO Studies.



investigator who likes to keep a low profile. Rodeghier asked him if he would look into what was going on. My friend said yes and immediately contacted Walsh at the *Muskegon Chronicle*. Walsh led him to Velthouse and the weather service operator. Velthouse was extremely cooperative. He verified his observation of one distant (and one

more distant) object through binoculars, gave angular directions, and approximated the angle above the horizon in a visit to the site. More about this important detail later.

The call to the weather station was less fruitful: There was no way they would immerse themselves in the media circus that was going on. My friend, however, didn't leave it at that. He called the NOAA regional office (the agency in charge of the NWS) in Silver Spring, Maryland, and described himself, CUFOs, and the investigation he was doing. On a hunch, he played the "college professor card," saying that a Western Michigan University science professor (me) would be involved. The response was noncommittal, and they told him to call back. They would have to think about his request to interview the radar operator.

A few days later, the return call was made, and the interview was on. NOAA had checked on the professor and agreed that I had to go with my friend. The interview would be taped (and they keep the tape). Appropriation reviews for NOAA were going on in Washington, and they didn't need any bad publicity. They wanted to cooperate, but wanted no surprises from us. They asked to receive any publications that resulted from the interview.

When we arrived for the interview, Muskegon Weather Service Director Leo Grenier and the young operator on duty during the UFO sightings couldn't have been friendlier and more helpful. (We agreed to keep the operator's name confidential.) We watched the operator demonstrate the capabilities of his radar station, and he was a real virtuoso. He pointed out signatures of planes and weather and indicated how utterly different the signals were on the night of the observations. Grenier looked on and agreed. There is no doubt that the operator saw something extremely unusual and unexplained.

We went to the conference room where the radar screen was simulated on an overhead projector with a standard overlay of the lakeshore region scanned by radar to the south. The sequence of events was vivid in the operator's mind. What had he seen? Although the returns varied in size and shape, many were kidney-shaped. He traced the pattern of the large kidney-shaped blips as they appeared to drift in the most puzzling dash-shift-and-hover motions out over Lake Michigan and, eventually, toward Chicago. The blips were immensely larger than planes, markedly sharper than weather returns, and unlike channeled reflections from atmospheric phenomena.

The position, size, and shape of the blips are visible in a series of nine drawings I made on radar charts during the interview. These are shown sequentially on the cover of this issue and on pages 4-7. (The weather-radar folks liked them so much that they copied a set for their files.) The blips never came near the radar site, and ranged mainly about 100-150 kilometers to the south. This put them south of Holland, which is consistent with the direction most witnesses saw the lights. Their altitude varied from 18,000 feet to more than 50,000 feet, and multiple blips were not always at the same height simultaneously.

Grenier and the operator specifically said that these returns were *not* a type of anomalous propagation because of their intensity and behavior, as well as the scope's ability to see all around, beyond, and beneath them. The intensity of the returns was great—"very reflective" being their exact words—or possibly self-emitting, that is, actively originating with whatever object was behind the returns. The radar sweeps produced very sharp aircraft-like spikes high in the air but much larger than normal aircraft returns.

DETAILS OF THE RADAR

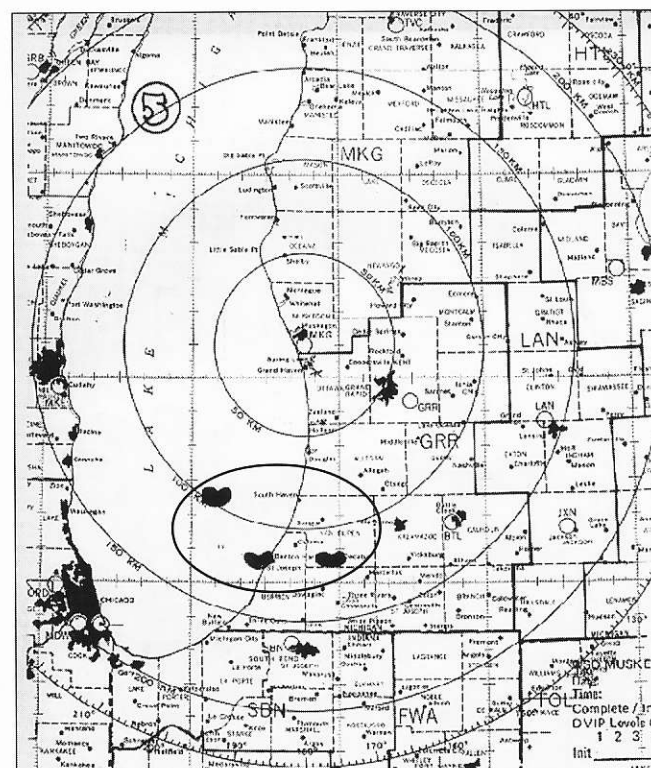
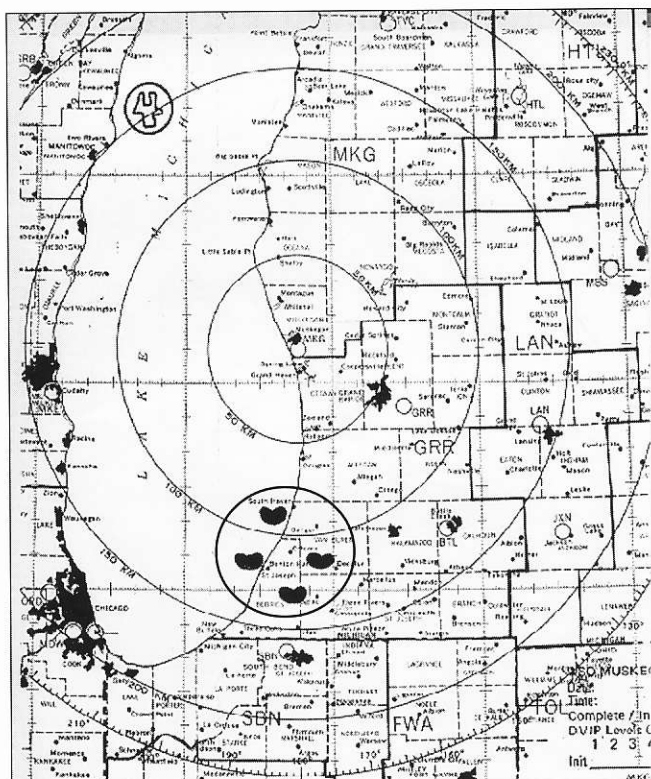
The weather station radar, a WSR-74C model (now being phased out in favor of more modern equipment), did not detect the unusual returns initially. Thus, the operator did not see anything upon first glancing at the sweep of the radar in its normal mode. The operator had to switch to manual mode and direct the beam south-southwest, back and forth in small angular sweeps or "paints." The blips then stood out clearly.

At South Haven, Michigan, the beam picked up the first blip at a distance of about 120 km (see Chart 1 on the cover). At that point, the beam was estimated to be about 4,000 feet wide during the sweeps. Therefore, in this manual mode, it didn't take much movement for the blip to be out of the beam (unless the object was exceptionally large). However, the operator was sweeping in a rapid manner (he guessed about once a second), and rarely lost the closest signal for any length of time.

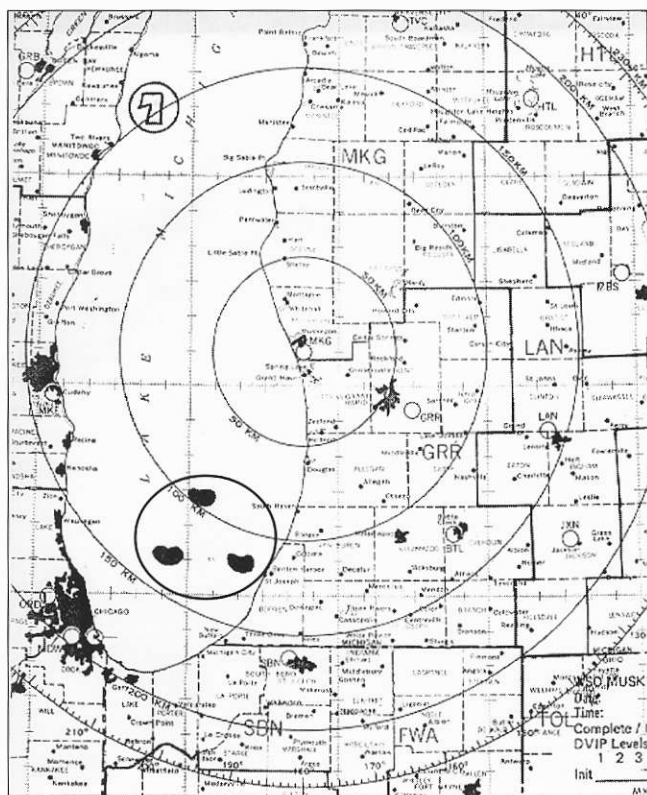
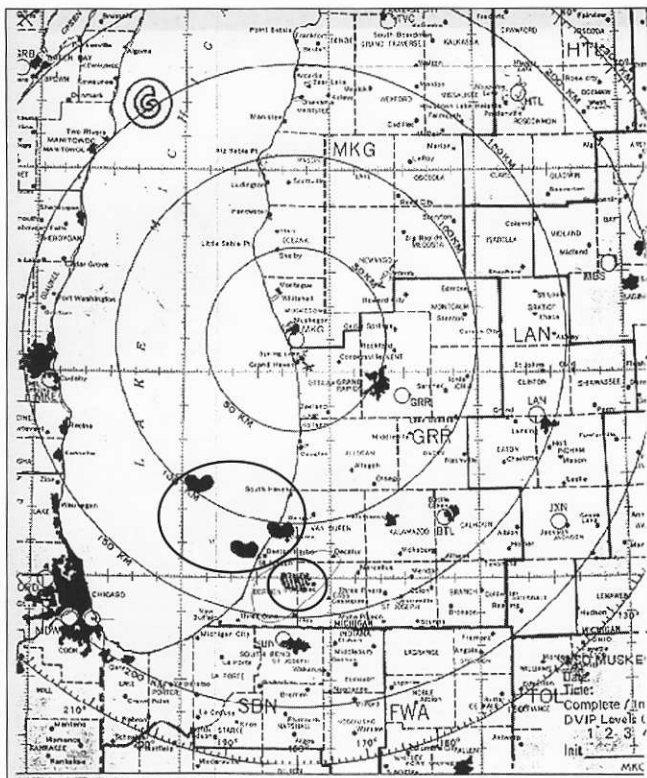
The size of the radar returns was unusually large. They were about the size of half of his thumbnail on the radar-scope, which covered a significant piece of air over the South Haven area. If the size of the blip related directly to the size of the UFO, then we would be dealing with some of the most gigantic UFOs ever. Of course, the blips from a plane are much larger than the actual plane, otherwise they wouldn't be visible on the scope; thus it is impossible to tell the size of the UFOs from the size of the blip alone.

The returns were much higher than reported in the media. This is because the operator in his amazement mistakenly said feet instead of meters during the call with the police. This is forgivable, as the charts list both measurement scales on the vertical axis, and feet is a natural unit of measurement for Americans. But instead of about 6,000 feet, the initial return was at 18,000 feet. And after a sudden ascent, it went not to 12,000 feet, but to around 40,000 feet, where, more or less, the rest of the dance was played out.

For the majority of the sequence (Charts 2–3, 5–8), there were three large, well-separated returns. These were sometimes in line, but mainly in a triangular array. Their movement was fascinating in that the closest blip would shift, almost instantly, leaving the others in place, and then the other two would shift, also almost instantly, to seemingly catch up and reform the triangle. An intermittent fourth signal blinked in and out. This sprint-and-drift shuffle sagged toward the southern end of Lake Michigan as time



passed, where eventually it dispersed into an uncountable ("three or four dozen?") number of dots in a small area (Chart 9), then disappeared. The weathermen had seen radar chaff many times on scopes before, and they said that none of this (even the last part) reminded them of anything like that.



AFTER THE INTERVIEW

After returning to Kalamazoo, my friend received the appropriate weather data for March 8 from the university, and he began a series of phone calls to NOAA in Silver Spring. He and NOAA came to agree that this was not

anomalous propagation or radar channeling. The weather was not right for the formation of a temperature inversion because of an offshore breeze, and the blips didn't behave like an inversion would. Thus, these were probably real returns of some kind.

My friend asked about the possibility of some military technology such as electronic countermeasures causing such effects, and the response was that this seemed unlikely due to the sharpness of the returns and the fact that people had seen something visually.

Attempts to work with the sheriff's office to locate new witnesses further south along the lake were less successful. There *were* such witnesses, including a retired USAF colonel who should have had a good look at the phenomenon, but these people were upset with the pushiness and lack of serious attitude they had witnessed, and wanted no part of any investigation. Their reticence hampered our ability to construct a continuous, correlated timeline—a very rare thing in ufology. The unfortunate gaffe that turned off the commercial pilot also hurt at this point.

Meanwhile, an odd piece of synchronicity occurred when a scientific colleague of mine informed me (following a church service, of all places) that UFOs had been seen that same evening by a professor (no names, please) and some students returning from a field trip. Their experience was of seeing four starlike objects (about 4th magnitude) spread across 10 degrees of arc in a line, smoothly moving from the direction of the lake in an easterly direction, over Kalamazoo toward Battle Creek. The group all agreed on this description. One student's boyfriend was a Western Michigan University pilot, and he too had seen the four lights from his airplane between Kalamazoo and Battle Creek slightly later, flying in an east-southeast direction. They were very high and technically smooth in their motion, as if connected (but not really connected, as stars were intermittently visible between one light and another). These Kalamazoo observations occurred about 15 minutes after the dissipation of the radar returns in southern Lake Michigan.

WHAT DO WE HAVE?

Due to various blunders and lack of coordination, it is very difficult to create a holistic picture of this important event. Here, however, is my best and most honest attempt:

1. Objects were seen sometime between 9:15 p.m. and 9:30 p.m. in the area of Holland, Michigan. The commonest description is of four lights, like a string of Christmas tree lights, which moved in sudden motions, sometimes like a ring or wreath, and oscillated in different colors (white, red, blue). These objects seemed to be high in altitude, but not at any great angle from the horizon (in other words, they tended to be more toward the horizon than overhead). The objects seemed to move from southeast of Holland to the southwest.

2. At about 9:30 p.m., calls began to come into the 911 desk at the sheriff's office and the Holland police depart-

ment, and three minutes later Officer Velthouse was on his way. In 10 more minutes, he was in the Graveses' neighborhood, and the phenomenon had moved off to the south-southwest. He viewed it in binoculars (one object nearer, one further). At the same time, the radar station began to paint its first blip in the same direction, though further away (at South Haven) than Velthouse thought it to be.

3. Nevertheless, the angle at which Officer Velthouse reported he saw the UFOs was very close to the expected angle that could be calculated from the radar data (see graph on the cover), indicating that this was a simultaneous radar-visual sighting.

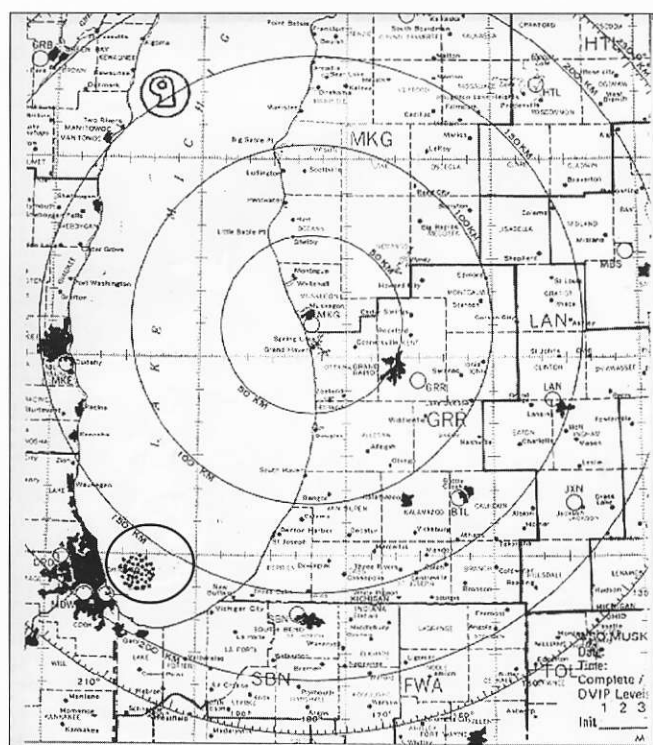
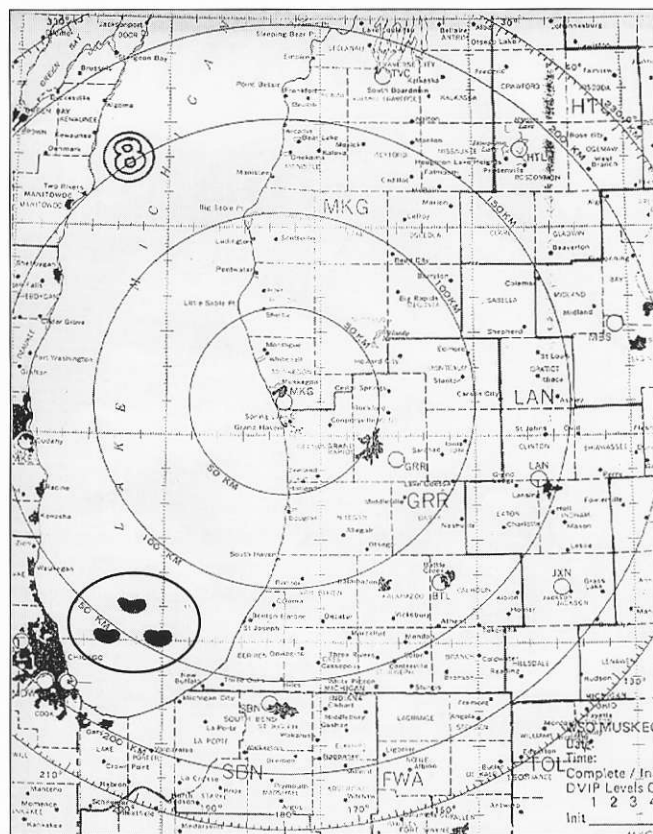
4. If these radar and visual sightings are truly correlated, the phenomenon continued for the next few minutes out into the lake and south towards Chicago, where it dissipated so far as the radar was concerned.

5. Approximately 15 minutes later, four smoothly moving, seemingly connected nocturnal lights were seen by several witnesses, both from the ground and in the air, moving easterly from the direction of the lake over Kalamazoo and toward Battle Creek. In numbers and activity, they were similar to the earliest Holland-area reports.

What remains as a large question is whether the Graveses' close-encounter spinning disk is the same phenomenon as the aerial lights, only up close. (They say it was, because they watched it from the disk phase until it was a distant light when Officer Velthouse arrived.) Also, the matter of how well the sightings of small lights at a distance were correlated with the radar returns remains to be determined. Time and distance seem generally good, but what explains the great size of the radar returns? Part of that size is just the characteristic way the radar set paints objects like planes much larger than they actually are. But still, these signals were far larger than those of airplanes. How so? Maybe we should leave *that* mystery to the ufonauts?

A WEIRD POSTSCRIPT

This anecdote doesn't add anything to the case, but it will nonetheless entertain you. Months after all this happened, a thin UFO newsletter came in the mail with the headline "Michigan '94 UFO Flap Unveiled! Nonexistent Documents Discovered!" Well, that was interesting. I opened the newsletter and got the sense of having stepped into a parallel reality of bizarreness. There, with a cloak-and-dagger commentary, was the story of the uncovering of secret NOAA documents concerning the case! And they were *my* hand-drawn maps I had done at the radar station in Muskegon! It took me about four pages to reestablish rational consciousness enough to say "Huh?" By a day or so later, I'd figured out my part in it. My friend reminded me that we'd left a copy with the radar folks. They had apparently passed them on to NOAA where they were filed. Upon being FOIA'd, back out they came. Wow! I didn't know how easy it was to create "secret documents." Well, the truth is now out: I've been rumbled.



I hope this report does something to reestablish everyone's consensus reality. And maybe, noting some of the blunders, it reminds us of how careful and patient we all need to be when interacting civilly with people when we investigate our cases. ♦

PROBING THE ROSWELL THIN-STRUT DEBRIS

BY ROBERT A. GALGANSKI

Most Roswell Incident skeptics dismiss the debris found on the Foster ranch as rawin radar-target wreckage from downed Project Mogul Flight 4. If they are correct, eyewitnesses must have misidentified slender “stiffener” fragments (broken glue-coated balsa-wood sticks) and irregularly shaped remnants of radar-reflecting sheeting (torn pieces of metalized paper) as highly unusual materials atypical of late 1940s technology.

Figure 1 shows a neoprene weather balloon and typical rawin target intact. Figure 2 depicts Roswell Army Air Field (RAAF) intelligence officer Major Jesse Marcel with the apparent remains of such a target. Some skeptics contend that Marcel, other RAAF personnel, and civilian eyewitnesses failed to recognize the stiffeners clearly visible in that photo because the dried-glue coating masked the balsa wood’s natural appearance and significantly changed some of its physical properties.

Two years ago, I conducted a series of straightforward home-workshop experiments with balsa wood that refuted—to my satisfaction, anyway—the Mogul explanation (see “The Glue Explanation Just Won’t Stick,” *IUR*, Winter 1997–1998). As a follow-up to my initial research, I interviewed on videotape debris eyewitness Jesse A. Marcel Jr., M.D. (the son of the late Major Marcel), on September 13, 1998, in Portsmouth, New Hampshire. I asked Marcel numerous questions about the debris he inspected as a boy and had him conduct on-camera experiments similar to those I had performed a year earlier.

This article covers selected portions of our dialogue. Various debris-related issues are addressed, including the possible material composition of the disputed thin-strut fragments. In this regard, Marcel’s recollections of the I-shaped piece he handled and other documented eyewitness

Robert A. Galganski has a master’s degree in civil engineering and has been researching the Roswell debris-field issue for nearly five years. A grant from the Fund for UFO Research permitted the original Marcel interview videotape to be edited for continuity and the inclusion of supplemental graphics.

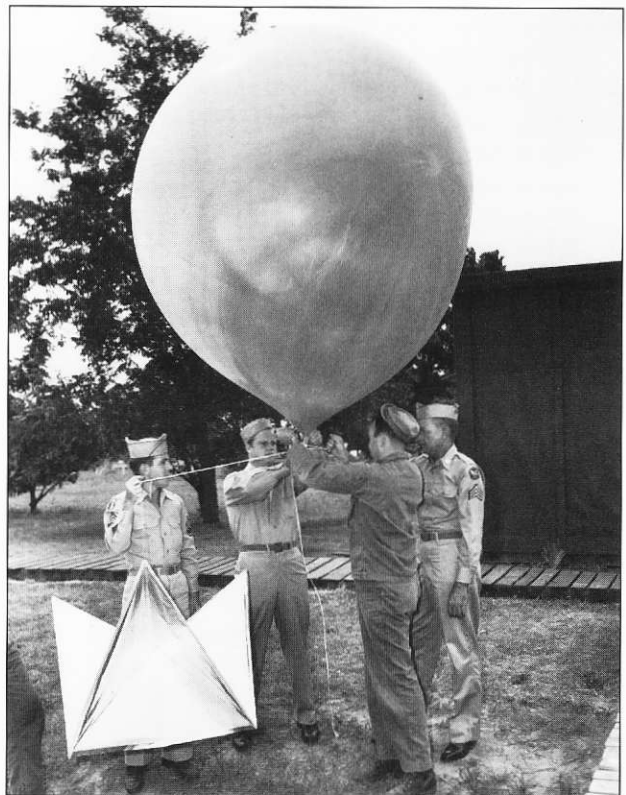


Figure 1. A neoprene weather balloon and a 1947-era rawin radar target.

observations of different-shaped, strut-like fragments are examined and discussed in detail. Their collective testimony is used to infer several key material mechanical characteristics that could explain the reported high bending resistance exhibited by these extremely lightweight, small cross-section structural elements. The conclusions drawn from this study also provide a rational basis for speculating about the type of object or device the struts may have come from.

The full version of the Marcel interview, including the results of the experiments he performed, is available in the *IUR* section of the CUFOS website (www.cufos.org/IUR_articles.html).

PHYSICAL PROPERTIES OF THE STRUTS

In early July 1947, 11-year-old Jesse A. Marcel Jr. reportedly handled a thin, capital-I shaped strut-like fragment with strange symbols on one web surface. The cover shows a 1995 replica (visual reproduction only) of that piece, the collaborative effort of Miller Johnson, industrial designer and artist, and Marcel. Its overall cross-section dimensions are $\frac{1}{8}$ -inch wide (across the flanges) by $\frac{3}{8}$ -inch high.

According to Roswell researcher Kevin Randle (private communication dated August 18, 1999), there are no other credible eyewitness descriptions of an I-shaped strut. Other reports of thin-strut pieces—from Major Jesse Marcel, Bill Brazel, and Loretta Proctor—mentioned a rectangular or some unspecified-shape cross section. These persons claim to have physically “tested” the struts or saw them being tested in some manner. If their accounts are taken at face value, the struts were extremely lightweight, slightly flexible but unbreakable when bent, resistant to slicing with a knife or other sharp-edged tool, and noncombustible when exposed to the flame from a cigarette lighter or match.

I prepared three $\frac{5}{16}$ -inch-square balsa-wood sticks—two of which had been coated with two different kinds of late 1940s-type glue—for Marcel’s inspection and use in the experiments. The dried-glue coating did not fool Marcel. He easily identified the core material as balsa wood, reiterating his oft-stated comparative assessment of what he had handled 51 years ago: “Metal is metal and wood is wood.”

Marcel pointed out after easily bending and breaking my balsa-wood test specimens that “it looks like a piece has been broken out of it. But this is not the way the I-beam looked. . . . it wasn’t shattered like that.” Figure 3 illustrates this important observation. The unmistakable signature of broken wood—notably, internal wood fibers sticking out of rough and irregularly contoured fracture surfaces—is clearly visible in that photo. That said, if the previously mentioned eyewitness reports were at least reasonably accurate, we can at least eliminate balsa—indeed, any—wood as a viable candidate material for the thin-strut debris. I would be shocked if the persons named earlier—an Army intelligence officer, a rancher, and a ranch homemaker, respectively—would not have been able to recognize, even after only a cursory inspection, broken wood. Nobody could be that sheltered or naïve.

BEYOND WOOD:

SOME STRUT SPECULATION

I asked Marcel a series of questions designed to gain some insight into the possible composition and function of the thin-strut fragments. By doing that, I implicitly assumed that his I-beam fragment and the



Figure 2. One of the July 8, 1947, Fort Worth Star-Telegram photos taken in Brigadier General Roger Ramey’s office showing a portion of what Roswell skeptics claim is the actual debris retrieved from the Foster ranch.

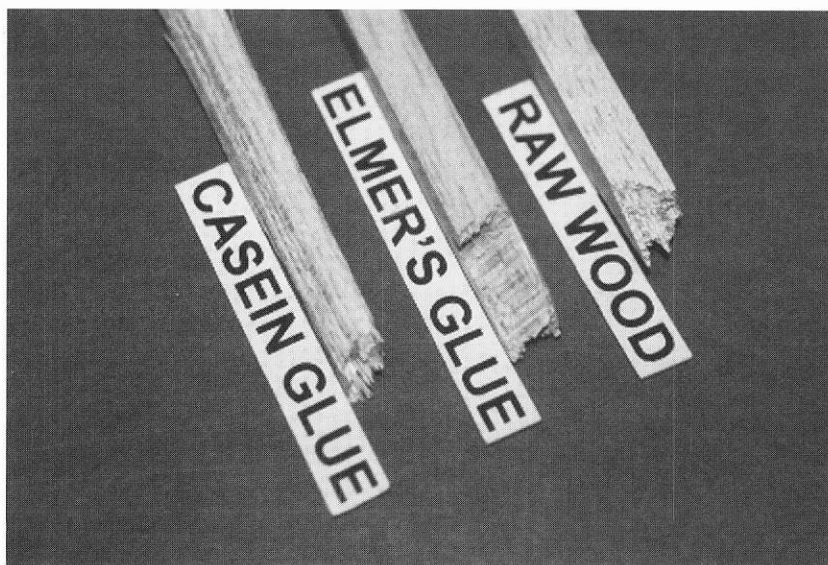


Figure 3. Closeup photo of a fractured end from each of the balsa-wood test specimens broken by Jesse Marcel Jr. Protruding wood fibers and “pockets” of pulled-out material are readily apparent.

different cross-section pieces reported by others were metallic and were made from the same material—a seemingly reasonable premise.

Question-and-answer highlights are presented below. Preliminary explanatory information is provided where necessary.

Could the struts have originated from an operational or towed airborne device? I did not ask Marcel this question directly. Rather, it reflects the end product of his response to one important question and a bit of deduction on my part.

Let's assume that the struts were part of a lightweight exterior shell structure from—perhaps attached to or encapsulating—a relatively small, operational aircraft, or a separate nonpowered, shell-like airborne device that was being towed by a flying craft. This premise is supported by the nature of the debris found at the Foster-ranch site. In addition to the struts, the only other kind of debris that could be regarded as having had a *structural* application consisted of a large—enormous according to some reports—quantity of thin-shell material. Wreckage commonly associated with an aircraft accident such as seats, electrical and electro-mechanical hardware, and propulsion-system components was conspicuously absent.

If the shell structure was an integral outer part of the craft, it's possible that the debris was generated after (1) a low-altitude, above-ground explosion, (2) a shallow-angle impact with the ground, or (3) a glancing impact followed by an explosion. In scenarios 2 and 3, the main part of the craft might have survived the accident, regained altitude, and resumed flight (perhaps to crash again at a second site). If the shell structure was being towed, it could have been either deliberately or accidentally dropped, with mechanical failure or an explosion as possible mechanisms in the latter case. (The first three scenarios are addressed in my article, "Roswell: Connecting the Debris Field and the Impact Site," *IUR*, Summer 1996.)

MATERIALS ENGINEERING 101

We'll further assume that the object or craft was *crash-worthy*. In other words, let's say it was designed to protect something of value—occupants, cargo, or instrumentation—in certain "survivable" impact scenarios. Crash-worthy vehicles use materials that have three common characteristics: light weight (a low density or weight per unit volume), moderate (as a minimum) stiffness, and ductility. It would be interesting to see how the strut's material properties measure up to these requirements.

The first material specification noted above—light weight—is self-explanatory; the others will require some discussion if you want to follow the arguments presented here. Stay with me.

Stiffness is simply the material's ability to resist elastic deformation. *Deformation* per se denotes a change in a material's original shape and/or size due to an applied loading or some other causative agent such as a temperature

gradient. *Elastic* deformation completely disappears when that loading or agent no longer acts on or affects it, provided the material is not deformed to such an extent that it exceeds its elastic-limit stress. (A very simplified definition of "stress" is the intensity of force per unit area of material.)

In the elastic range, a material's stiffness is measured by its modulus of elasticity; the higher the modulus, the stiffer the material. To give you a feel for what this parameter indicates, common structural steel is about three times as stiff—therefore three times more resistant to deformation—as ordinary aluminum and aluminum alloys. Steel is roughly 30 times as stiff as all woods. Aluminum and its alloys are about 10 times stiffer than wood.

Two things can happen to a material that is stressed beyond its elastic or self-restoring deformation region. It will either fracture—physically break apart—or become permanently distorted. This difference is extremely important from a design standpoint.

Those that fracture are called "brittle" materials. They never experience any appreciable amount of permanent deformation, regardless of their stiffness. Typical examples include wood, ceramics, cast iron, concrete, certain metallic alloys, and glass.

Other materials don't break immediately when stressed beyond the elastic range. However, they will not regain their original size or shape once they're stress free (when the loading or other stress-causing agent is removed). Engineers say that such "ductile" materials "yield" or are stressed into their so-called plastic range of deformation. Some materials are more ductile than others. Low-carbon (structural) steel, many aluminum alloys, and polyethylene are examples of materials that have especially high ductility.

High-ductility materials can withstand *extensive*—and often massive—permanent deformation before fracturing. They can handle loads significantly higher than their design counterparts (which are always conservative) because of their ability to "flow" somewhat before breaking. Brittle materials can't do that. They just break. Period.

For this and other reasons that would occupy too much space to explain, ductile materials are employed for use as load-carrying structural elements in vehicles of all kinds: ground, water, and air. They can withstand extremely high impact loadings that stress a material far into its plastic range of deformation. Although vehicle structures are *expected* to collapse in even relatively moderate-severity impact scenarios, they must do so in a *controlled* manner, with minimal fracture failures. This structural integrity enables their individual components to dissipate the maximum amount of kinetic energy possible—a critical design objective. Material ductility alone isn't good enough for such applications. Materials having sufficient "toughness"—an adequate combination of stiffness *and* ductility—are required. Brittle materials, which break quickly without dissipating very much kinetic energy, obviously need not apply.

THE STRUT SUBSTANCE

If I still have any readers left at this point, let's get back to Roswell. All persons who held thin-strut fragments from the debris field said they were extremely lightweight, consistent with the first material specification mentioned at the beginning of this discussion.

Major Jesse Marcel and Bill Brazel reported that they had experienced great difficulty when they tried to bend thin-strut pieces found at the debris-field site. Neither one apparently could generate the force required to bend them enough to either break them (if they were brittle) or cause them to yield (if they were ductile). From elementary mechanics of materials theory and personal experience, this indicates that the strut bending resistance—as measured by the product of its modulus of elasticity and a calculated cross-section area section property called the moment of inertia—must have been *at least* roughly comparable to that of a $\frac{3}{8}$ -inch-diameter aluminum rod. (I could not bend such a member by hand, no matter how far apart on it I placed my two hands.) From this we can conclude that the struts appear to have been made from a material that had, as a minimum, moderate stiffness—certainly much higher than that of brittle balsa wood (see the material relative-stiffness examples provided earlier). That satisfies the second material requirement.

Unfortunately, Major Marcel's and Brazel's anecdotal "test data" can't tell us if the strut material was ductile, the third material spec in this discussion.

But we *can* use well-known facts about how newly fractured material surfaces look to indirectly try to answer this question based on admittedly scanty "evidence." In general, the freshly broken surfaces of most ductile metals—high-tech or not—have a dull, smooth, and velvety appearance. For brittle metals, those surfaces will be bright, rough, and granular.

These surface-texture signatures have been observed in tests performed with individual components under controlled laboratory conditions. Determining or verifying general material type this way when the same components are part of an assembly or an entire structure that has failed in its actual operating environment can, however, lead to an erroneous evaluation. Numerous factors such as load mode and duration, stress type, and operating/ambient temperature, complicate the process. For example, some ductile materials may exhibit—totally or in part—brittle-material fracture-surface characteristics when they are subjected to *extremely rapid* shock loads and don't have time to start stretching. Lacking any information whatsoever about how the debris-field struts broke, I had to assume they failed as if they were lab tested, without the presence of any extreme complicating factors.

When I asked Marcel how the assumed broken ends of the I-beam fragment looked relative to the two surface-texture choices, he selected the first group of descriptors and added: "I don't remember a big difference between the

ends of it and the way it looked . . ." If his recollection is accurate and if the conditions surrounding the strut failure were not unduly complicated, we can conclude that they were made from a ductile material.

The struts therefore appear to satisfy all three material specifications mandated for crashworthy vehicle design. As such we can conclude that they *could* have been part of a powered aircraft or a towed airborne device that suffered some mishap above the Foster ranch. This finding is, of course, speculative. But it is based on logical conjecture supported by sound engineering principles.

A material laminate? George Filer, UFO researcher and director of advanced technological exhibits for the Air Victory Museum in Medford, New Jersey, requested that I ask Marcel about the possibility that the I-beam was a laminate comprising more than one lightweight material. ("Laminate" is a generic term denoting a structural element made from at least two layers of bonded material.) Filer said that the Germans and U.S. aviation companies started incorporating such materials into experimental aircraft during World War II. He noted, for example, that Chance Vought combined aluminum and balsa wood in selected structural elements used in their V-173 Flying Flapjack, a pancake-shaped fighter prototype built for the U.S. Navy and first flown in 1942.

Filer speculates that whatever (presumably) crashed or exploded at the Foster ranch was a top-secret U.S. aircraft that was fabricated, at least in part, using such a material laminate.

When asked if this is what he saw back in 1947, Marcel stated that the I-beam fragment cross section "was *uniform* as far as I could tell" (emphasis added). His reference to cross-section homogeneity appears to rule out the possibility that the I-beam and (as per our assumption) the rectangular struts were merely laminated structural "sandwiches" from a 1940s-era advanced aircraft.

Was there a residue on the debris, and did the debris have a distinctive odor? A crash or explosion involving a conventional or experimental aircraft could have left traces of the mishap on some or all of the debris that fell to the ground. The evidence could have been in the form of a residue such as meltage or powder from one or more mundane substances such as plastic, rubber, or fabric on its surfaces; discoloration and warping due to the presence of intense heat; or impact-induced gouges, scratches, or other indentations. Some of these processes or events could also have caused the debris to have a distinctive odor.

When asked about some of these possibilities Marcel replied that all of it "looked clean. There was no dust or particulate matter embedded on the foil, Bakelite, or the beams." Regarding the I-beam fragment in particular, he remarked, "There was no glue or any coating on this. When you handled it, when you were touching the surface, there was no additional coating . . . no splotches, no grain, no nothing. It was a piece of metal—uniform."

Marcel did not remember any particular smell emanat-

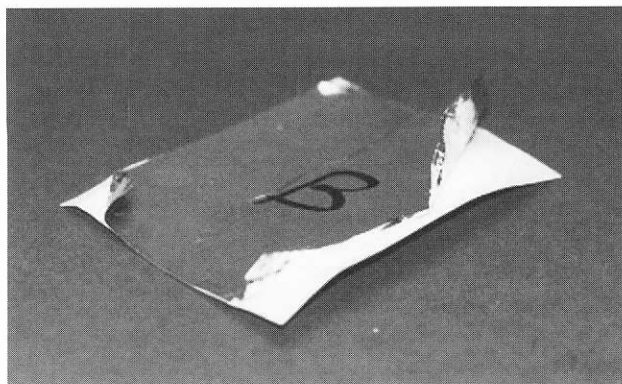


Figure 4. A partially delaminated piece of simulated rawin radar-target laminate made from aluminum foil and paper. Portions of similar materials depicted in the Fort Worth, Texas, photos had also separated from each other. Most of that damage probably occurred when the target impacted the ground and was subsequently dragged over vegetation and exposed rocks by the remaining inflated balloons in the Mogul Flight 4 or equivalent balloon train.

ing from any of the debris he inspected.

At first glance, Marcel's responses to these questions seem to preclude the possibility that the Foster-ranch debris originated as a result of an aircraft accident. However, it's also possible that the violent thunderstorm reported in the area on the night of the alleged crash or explosion may have deposited a significant amount of rain on the debris, washing away any residue (and associated odor, if any) that may have been on it. Moreover, since he saw or handled only a very small portion of that material, it may not have been representative of the vast majority of stuff on the ground.

The best we can say is that Marcel's recollection of no residue or odor isn't much of a test and the above-noted questions cannot be answered with any degree of certainty.

MARCEL EXAMINES THE "DEBRIS" PHOTOS

Even though he probably saw them before, I documented on videotape Marcel's reaction to and comments about a few of the alleged Foster-ranch debris photos taken in Brig. Gen. Roger Ramey's Fort Worth, Texas, office on July 8, 1947. Roswell researcher Bob Durant provided me with 16-by-20-inch prints showing various views of the disputed materials; I took four of them with me to Portsmouth for use during the interview. Some noteworthy remarks Marcel made about their contents are summarized below.

Marcel identified the stuff on the floor of General Ramey's office as balsa-wood sticks and "metalized paper" (the aluminum foil-paper laminate) from a rawin target, the remains of a balloon envelope, and some wrapping paper. According to him *none* of it looked like the debris he remembered seeing and handling as an 11-year-old boy.

I asked Marcel if he distinctly recalled looking at both sides of the thin, shell-like material back in 1947. He did

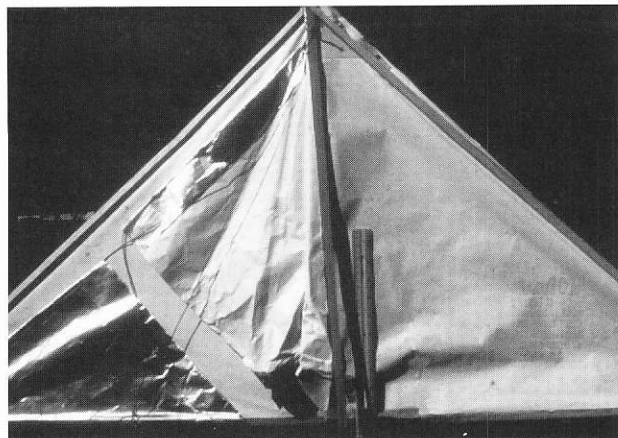


Figure 5. Overall view of a folded 1951-1953 rawin radar target provided by U.S. Air Force Captain James McAndrew to UFO researcher Rob Swiatek for his inspection. Photograph by Rob Swiatek.

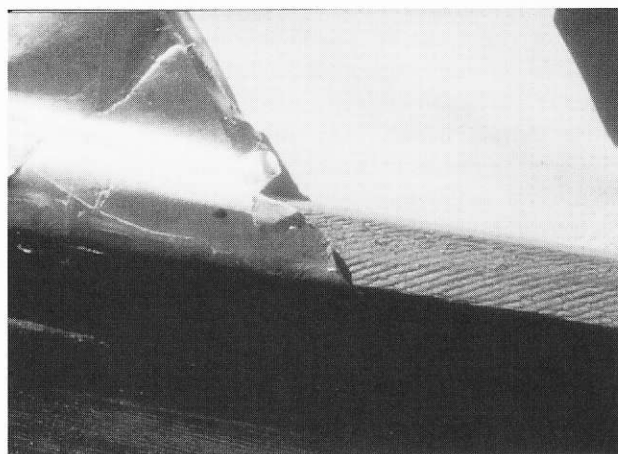


Figure 6. Closeup view showing how the metalized paper sheeting was wrapped around the wooden stiffeners comprising a 1951-1953 version rawin target. Photograph by Rob Swiatek.

and asserted they were entirely metal, completely devoid of paper. Then he added—adamantly—"There's no question in my mind about that."

In an effort to calibrate the significance of that statement I asked Marcel to identify the constituent materials in a piece of rawin-target laminate I had made from heavy-duty aluminum foil bonded to thick, white paper with Elmer's Glue-All. He had no problem at all, responding immediately: "Metalized paper." I then asked him to separate the two layers of material and hold the partially delaminated piece shown in Figure 4 next to one of the "debris" photos. My homemade "damaged" fragment was a close match, at least visually, with the radar-target reflective-material scraps shown there. The aluminum and paper parts of both the original and simulated laminates stood out like hammer-caressed thumbs.

(continued on page 30)

ASTRONOMERS AND UFOs

BY HENRY CORDOVA

Every year, in late summer, the astronomy department of the southern university where I was a student holds its open house. One particular open house in 1971 was especially noteworthy. The department was showing off its brand-new astrometric reflector and though there were still a few bugs to be worked out before the big telescope was fully operational, it was working fine for visual observations. Dozens of guests were lined up for breathtaking views of the planets, bright clusters, and nebulae.

Unfortunately, the weather was not cooperating. It was about 50% overcast, and low-flying scud clouds interfered with spectacular views of a very dark and transparent sky beyond. It was one of those situations familiar to any astronomer: If it cleared up just a little, some good observations were still possible; if it got any worse, we would have to pack it in and go home.

Fortunately, the guests still had a lot to do. The graduate students were manning a battery of six- and eight-inch reflectors in the parking lot, and guided tours of the observatory facility were in progress. Besides the telescope, there were other things to see: the plate archive, the measuring engine, and the refreshments.

"Dr. Davis" (pseudonyms are used here to avoid embarrassment) was well-liked by both staff and students; he was first-rate in his field and had a reputation as a snappy dresser. He was handling his position as master of ceremonies with the panache of a game-show host. His goatee was perfect.

Except for the weather, things were going well. As an undergraduate helping out with open house, I was excited to be there. Observatories are holy places, and I was eager to communicate this enthusiasm to our guests. Not present at our gathering was the chairman of the university's astronomy department, "Dr. Klein." Whether he had cancelled his appearance due to other commitments or simply had not yet arrived I can't recall, but I do know that he was not present during the events that followed.

Klein was an unforgettable man of vast proportions. He was enormous in both height and width, and he could shock an entire auditorium into silence by simply walking in.

Henry Cordova is an amateur astronomer who lives in Tamarac, Florida.

His Teutonic-English accent could be forgiven and he spoke several languages besides his own, perfectly filling the stereotype of the European renaissance man. He seemed to know everything, and besides being one of the top men in his own branch of astronomy, he could carry on a debate with anyone on any subject. At least, his students thought so. Klein treated us fairly, but we were nonetheless terrified of him.

He was also a well-known figure in the community. His pet peeves were UFOs and astrology. Every time either subject appeared in the local news, he went on the warpath with letters to the editor and interviews on the local radio and television shows. Although his staff and students all agreed that this nonsense required vigorous rebuttal from legitimate science, we often wondered if his shrill opposition might not have just the opposite effect.

With the observatory roof fully rolled back while the big scope was being used, Davis must have heard the commotion in the parking lot from inside the building. He came running out of the building where his dark-adapted eye caught the entire crowd frozen in a heart-gripping tableau. Everyone was pointing up at the sky, yelling "UFO! UFO!"

It was indeed a classic UFO. A featureless, single, blue-white light, brighter than Venus, was traveling across the sky behind the clouds. It was not an airplane. I saw it through binoculars and could see no navigation lights. Neither could an alert student who managed to get a brief glimpse of it in the field of a six-inch reflector. It was not a satellite; these were all several magnitudes fainter. Besides, its orbit was from southeast to northwest, and it never faded into the earth's shadow. It finally disappeared in the far west. Still the real show was not in the sky, it was in the parking lot. Everyone was excited and pointing, and Davis, who was standing on the steps of the building just beyond the front door, got a great look at it for a while. Then he spun around and walked into the observatory. The pandemonium in the parking lot continued until well after the object disappeared.

It is not necessary to assume an extraordinary cause for an unidentified light in the sky. In spite of the public reaction, I have no doubt that the mysterious light had a perfectly reasonable explanation. An unusual light display on an aircraft or satellite is always a possibility, as well as

the usual suspects—illuminated weather balloon, a deliberate hoax balloon, perhaps even a reflection from an inversion. Not that it matters; one weird light does not an alien spacecraft make.

The truly extraordinary phenomenon here is the reaction of the senior scientist present. Davis could have conducted interviews, passed out questionnaires, surveyed the audience (at least the experienced observers present in the audience, about half a dozen astronomy students). What a marvelous opportunity to debunk the lay myth that science refuses to even look at evidence challenging the

current orthodoxy! Alternatively, it might have been possible to identify the mysterious object by publishing our observation and soliciting additional information through the press. I can only speculate on what was going on in Davis's mind. Perhaps he had a fleeting nightmare vision of television cameras and microphones, the morning edition and film at eleven. Or even worse, the next staff meeting.

That night, and over the next few days, Davis refused to discuss the incident. When confronted, he either changed the subject or left the room. At any rate, Klein never found out about it, and I wasn't about to tell him either.

ASTRONOMERS AND UFOs: AN EPILOGUE

BY FRANK JOHN REID

Three stellar ufologists—real scientists who studied ufology in the same era as Henry Cordova's observation above—each came across cases of similar deliberate ignorance in astronomers. Astronomy (the original home of the term "extraterrestrial") has always been willy-nilly the science called upon to judge "flying saucers."

J. Allen Hynek, the astrophysicist who founded the Center for UFO Studies in 1973, began the first chapter of his *The UFO Experience: A Scientific Inquiry* (1972) with these words:

During an evening reception of several hundred astronomers at Victoria, British Columbia, in the summer of 1968, word spread that just outside the hall strangely maneuvering lights—UFOs—had been spotted. The news was met by casual banter and the giggling sound that often accompanies an embarrassing situation. Not one astronomer ventured outside in the summer night to see for himself.

Hynek went on to say he had many times watched reactions when the UFO topic came up in gatherings of scientists:

Frequently the reaction has been akin to that of a group of preteenagers watching a movie scene of exceptional tenderness or pathos quite beyond their years to appreciate: giggles and squirming suggest a defense against something the scientists cannot yet understand. . . . Perhaps they are expressions of deep-seated uncertainty or fear.

In a statement prepared for the U.S. House of Representatives' Committee on Science and Astronautics (*Symposium on Unidentified Flying Objects*, hearings, July 29, 1968), James E. McDonald, meteorologist and atmospheric physicist, addressed the question: Why don't astronomers ever see UFOs?

I have had this question put to me by many persons, including a number of astronomers. Once I was speaking to a group from an important laboratory of astronomy when the director asked why astronomers never see them. In the room, among his staff, were two astronomers who had seen unconventional objects while doing observing but who had asked that the information they had given me about their sightings be kept confidential. I understand such strictures, but some of them make things a bit difficult.

McDonald went on to relate some known sightings by astronomers, and remarked, "It is often quite difficult to persuade a scientist to let his confidential report of a UFO sighting become a fully open UFO report; and my own experience suggests that perhaps astronomers, as a group, are just a bit more sensitive on this score than other scientists."

Jacques Vallee, the astronomer and computer scientist who has a real claim to being one of the early originators of the Internet, experienced the most egregious—or the most pathetic, or the most puzzling—of this sort of behavior. In June 1961, young Vallee was hired for the Paris Observatory's artificial-satellite tracking program, headed by Paul Muller. The trackers used theodolites (small, mobile telescopes with scales to show exact elevation angle, and exact direction in degrees clockwise from true north), a precision chronograph, and a tape recorder. Set up to anticipate satellite passages, they would call out observational data for use in refining orbits. This data would be telexed to the world center for satellite tracking at the Smithsonian Institution in Washington, D.C.

Sometimes they saw things that were unspectacular but quite unanticipated, as Vallee recorded in his diary (published in 1992 as *Forbidden Science*):

Paris. 12 August 1961. . . .

Occasionally we observe objects that remain uni-

Frank John Reid is historical consultant for the Center for UFO Studies. He lives in Chicago.

identified. Thus on 11 July at 10:35 p.m. I saw a satellite brighter than second magnitude. I had time to log a few data points. On another occasion several of us recorded no less than eleven points. The next morning Muller, who behaves like a petty Army officer, simply confiscated the tape and destroyed it, although a similar object had just been tracked by other astronomers at Besançon and by Pierre Neirinck, a satellite expert based in Saint-Malo.

"Why don't we send the data to the Americans?" I asked him.

Muller just shrugged.

"The Americans would laugh at us."

He seems terrified at the idea that the morning papers might come out with the headline: "Paris Observatory tracking something it cannot identify." Muller is a tough man who believes in discipline and a simple world where everything is neatly labeled. His previous career in astronomy was based on measuring the angular separation of thousands of double stars, the most painstaking work imaginable. He approaches artificial satellites in the same spirit.

Later on, when Aimé Michel (much worried about his health) offered to donate his UFO materials to the observatory for preservation, Muller had the brass castors to say: "You see, that's another letter for the crackpot file. Although properly speaking Aimé Michel is not really a crackpot, he is a crook."

To offer a parable: Galileo didn't invent the telescope, but he was first to apply it to astronomy. Largely on its

evidence, he asserted Copernicus's heliocentric planetary system (including the earth) to be not just a useful mathematical model, but the proven literal truth.

It's a historical commonplace that a number of Galileo's astronomer critics refused to look through the telescope. This is cited as a great folly. But they had a point: How did Galileo know that telescopic optics, which worked well enough over a few terrestrial miles, remained constant over planetary distances orders of magnitude greater? Could they be sure they wouldn't see optical illusions?

The folly was not in skepticism, but in not letting the observations by the better observers lead them into the hard, grubby work of testing the instrument, or at least promoting the testing by others. (It's well-known that you have to *learn to see* when you first use a telescope, even modern ones.) To use a theological term, it's intellectual sloth.

The *Skeptical Inquirer* for Sept./Oct. 1999 quoted physicist Robert L. Park: "Alas, to wear the mantle of Galileo it is not enough that you be persecuted by an unkind establishment, you must also be right."

But I think Galileo owned two mantles. One was glorious, of purple velvet lined with cloth of gold, and he wore it—and bequeathed it to his successors to wear—only on being wreathed with victor's laurel by history. The other was of musty, worn, tenuous broadcloth, with which he kept out the chill after the curiously fastidious amongst his peers refused to look through the telescope.

At CUFOS, we wonder exactly how widespread this sort of behavior by scientists has been. Do readers have any more examples of it? And does it still occur?

ASTRONOMERS AND UFOs: SOME OTHER EXAMPLES

BY GEORGE M. EBERHART

Before the UFO phenomenon became widely identified as disreputable pseudoscience, astronomers were able to report strange occurrences in scientific journals without being stigmatized. The only requirement was that the observations be detailed, accurate, and instrumented, if possible.

British astronomer W. H. Stevenson described "bright objects observed near the sun" on September 9, 1914, in a communication to the *Journal of the British Astronomical Association*. Hundreds of round or dumbbell-shaped objects passed across his telescopic view singly, about once every five seconds, for three hours. Whether these were really airborne seeds, insects, or other aerial detritus is moot; if Stevenson had reported his observation in 1999,

it would have been labeled a UFO sighting. But in 1914, he could be open-minded:

Though the phenomenon described in this paper was probably not astronomical but purely terrestrial in origin, the excuses pleaded by the writer in bringing the matter forward are that the appearances were encountered in the ordinary course of observation; that similar observations have from time to time been reported by other observers, especially within recent months; and that a satisfactory explanation of them is still lacking.

This type of observational objectivity, though rare, has persisted into the latter half of the 20th century. A notice in the *Strolling Astronomer* for 1955 under the heading "An

George M. Eberhart is director of publications for the Center for UFO Studies.

(continued on page 32)

FIRE ON THE MOUNTAIN: THE BERWYN MOUNTAIN INCIDENT

BY ANDY ROBERTS

There was no doubt in my mind that something of great significance had occurred on the Berwyns. ... I leaned towards accepting the extraterrestrial hypothesis.”—Nick Redfern, in *Cosmic Crashes* (1999)

In his novel *By Spaceship to the Moon* (Basil Blackwell, 1958), Gavin Gibbons imagined a UFO landing on the Berwyn Mountains in Wales. Sixteen years later, in 1974, those same mountains would again be the focus for a story involving a downed UFO. But this time, some said, the story was for real. The Berwyn Mountains run southwest to northeast across central North Wales, separating Shropshire from the Snowdonia National Park. They have a long history of human habitation. Prehistoric peoples lived and worshipped on the mountains, leaving behind a dramatic ritual landscape to which many strange beliefs have become attached. UFOs are not new to the area, and local folklore tells us that these peaks have been haunted by a multitude of aerial phenomena, including some spectral hounds of hell, while to the south at Llanrhaiadr-ym-Mochnant the villagers were once plagued by “flying dragons”—a common historical name, some have theorized, for UFOs.

Contemporary enigmas abound as well. These include “phantom bombers,” ghosts, and lake monsters. The region is also the lair of that most modern of mysteries, the “alien big cat.”

Though popular as a tourist destination, the Berwyn Mountains can be treacherous, and mountain rescue teams are frequently called out to search for the lost and injured. The highest peak, Cadair Berwyn, rises to 827 meters. Several airplanes, both military and civilian, have crashed on its slopes in poor visibility over the past 50 years. In winter the area is especially remote, often snow-covered, and dark for more than 12 hours a day—an ideal spot, if ever there was one, for a UFO landing.

Andy Roberts has been a UFO investigator for 25 years. Besides writing books and articles dealing with UFOs, he has also published in the Fortean field. In the real world he works at a hostel for homeless ex-offenders and harbors a fascination for the Grateful Dead.

RUMOR AND SECRECY

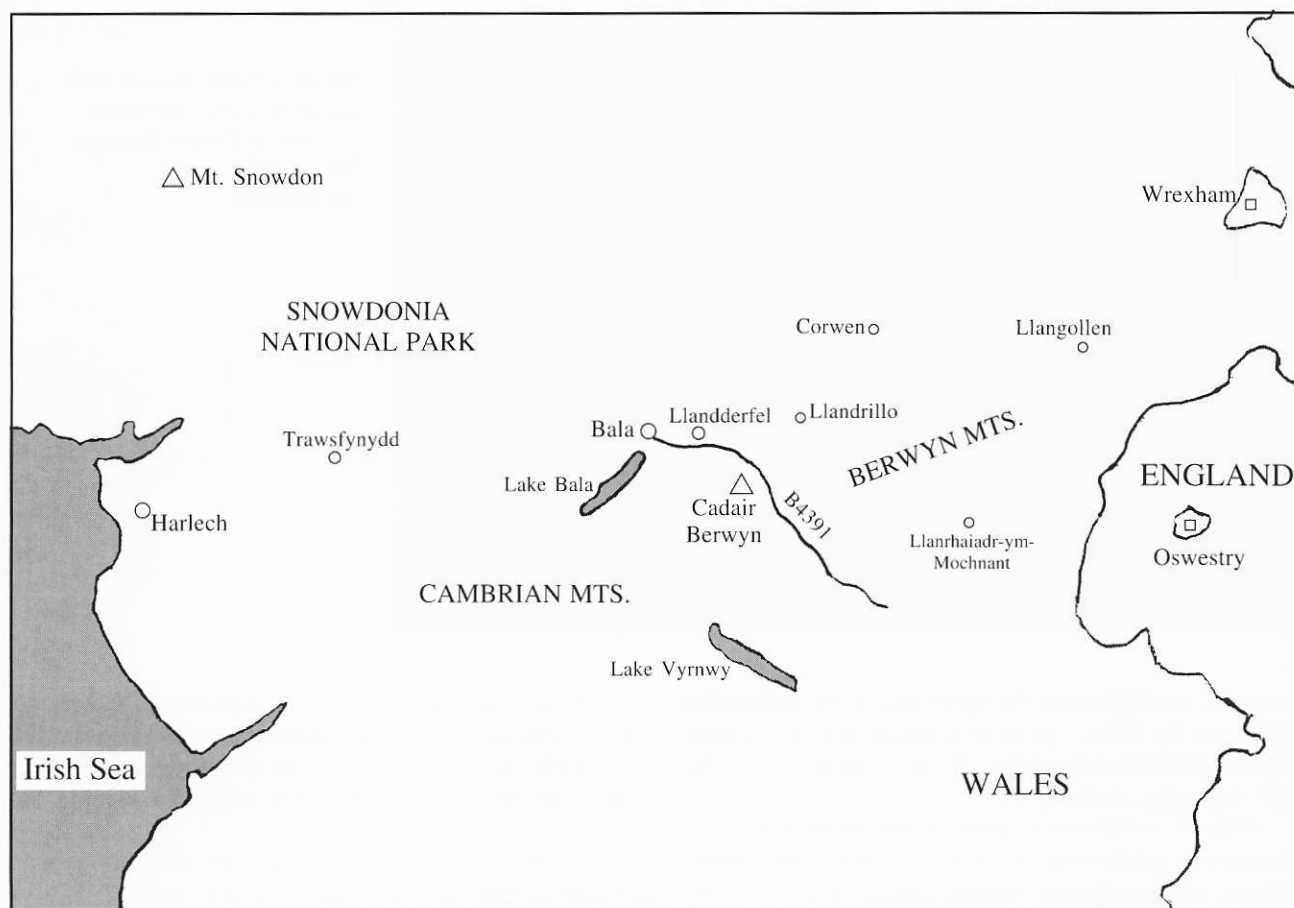
It is against this backdrop that an incident took place on January 23, 1974, an event that perplexed locals and later the UFO research community. The event spawned a cascade of rumors that led some investigators to conclude that an extraterrestrial craft had crashed on Cadair Berwyn. These same ufologists also claimed that the alien crew, some still alive, were immediately whisked off to a secret military installation in the south of England for study and that the whole fantastic business has been hushed up by the British government. The Berwyn Mountain incident has been described as “the best example of a UFO retrieval in Britain” and likened to the Roswell and Rendlesham events.

A preposterous claim? Certainly, and a claim easily dismissed by those with little or no knowledge of the case. But there is no smoke without fire. Even the most outlandish story must have its genesis in some truth.

If there was no alien craft, what lies behind the longevity and tenacity of these claims? Could it have been the crash of a secret military test craft such as one of the “flying triangles” that have attracted both ufological and popular attention throughout the 1990s? Or perhaps a failed missile test from the rocketry range at nearby Aberporth? A hoax even? Or something far more complicated? And if it is any of these, why have the claims of UFOs, alien cadavers, and military cover-ups persisted for a quarter of a century?

Comparisons with Roswell and other UFO crash-retrieval claims show the Berwyn Mountain incident to have many of the same components and motifs and therefore to be worthy of an in-depth examination.

Yet, while rumors of this crash have been in existence for 25 years, it has only recently drawn any serious attention from the UFO community. Although dramatic allegations have been made, no one had investigated this potentially remarkable case in any great depth. The incident, far from proven, became a kaleidoscope of lore and fact about bodies, military retrieval teams, earth tremors, meteorites, weapons-testing, disinformation agents, men in black, and geologically created lights.



THE INCIDENT

The story is a complex one. I have pieced together a composite account from statements and articles by witnesses, informants, ufologists, and newspapers of what allegedly happened on and around January 23, 1974. This is the generally accepted account, variations of which have become enshrined in the UFO literature and seeped out into the public's consciousness.

Prior to the Berwyn Mountain incident, the north of England had been plagued by an aerial phenomenon dubbed the "phantom helicopter." More than 100 good sightings were made of this anomalous object, seen flying low at night often over dangerous terrain and in appalling weather. These sightings largely took place between spring 1973 and spring 1974 and ceased, coincidentally or curiously, immediately after the Berwyn Mountain incident. Despite the numerous sightings and keen police interest, which led to a still-secret official report, no one explanation was ever found. But something was flying around the northern skies, and many of the witnesses concurred that whatever it was, "it seemed to be looking for something."¹

That Wednesday was just another day in Bala and the nearby villages of Corwen, Llandrillo, and Llandderfel. UFOs were the last thing on the villagers' minds as Britain shivered in midwinter and the recently introduced three-day week. But as night closed in, something odd occurred.

Just after 8:30 p.m., thousands of people in the area were jolted from their winter musings by at least one, or possibly two, explosions, followed immediately by a terrible rumbling. The event lasted four or five seconds. Furniture moved, ornaments rattled, buildings shook. Livestock and domestic animals voiced their terror. As people shot to their windows, some saw lights streaking across the sky.

Villagers flooded out into the streets in an attempt to discern the cause of the violent disturbance. As they looked up into the mountains, several saw a mysterious white glow, lasting a few seconds. Others saw beams of light being projected into the night sky.

Many villagers immediately called the emergency services, believing that a disaster of some kind had taken place. After speaking with the police, one local nurse was certain that an aircraft had crashed. She set off for the mountains in her car, dreading what she might find there but determined to offer help until the emergency services arrived. Once above the tree line and on the high mountain road, she stopped her car, baffled and startled at what she could see. There, high on the desolate mountainside, was what appeared to be a large, glowing sphere. Whatever it was, it lay too far from the road to be reached on foot. All the nurse could do was watch. The sphere seemed to pulsate, changing color from red to yellow to white, while other white lights—"fairy lights" as the witness described them—could be seen above and below it on the hillside. Realizing she



The spiky rocks of Foel Wen add foreground interest in this view of Cadair Berwyn. Photo credit: Lou Johnson

could not possibly reach the lights, she drove back to her village. As she did so, a group of police and soldiers stopped her and forcefully ordered her off the mountain, saying the road was being cordoned off.²

Official reaction was quick to the initial explosion; suspiciously quick, some say, with more police and military arriving within minutes, turning people away from the mountain roads. In the days following, there was an unusually large military presence in the area. Roads remained closed and farmers reported they were forbidden to tend their stock. Obviously, someone was looking for something, or why would military jets and helicopters criss-cross the area and strangers comb the mountainside? Scientists from universities also came to tramp the hills, but far more suspicious were the official-looking outsiders who turned up in the villages immediately after the event, tight-lipped about their business but keenly interested in the events on the mountain.³

The incident was immediately taken seriously by the media, with national television and radio reports being broadcast over several days. *The Daily Telegraph*, *The Guardian*, *The Times*, and other national newspapers gave the event considerable coverage, as did the Welsh regional and local press.

Speculation about the cause of the explosion, rumbling, and lights was rife.⁴ An aircraft crash would have accounted for the noise, lights, and keen official involvement. Indeed, one local newspaper was certain that whatever had taken place involved a crash of some kind and that something had been retrieved from the mountains. It noted, "There is a report that an Army vehicle was seen coming down the mountain near Bala Lake with a large square box on the back of it and accompanied by outriders."⁵ But the authorities steadfastly refused to acknowledge that anything unusual had taken place.

In any case, not one of the "explanations" took into account the totality of what had been reported by witnesses. Meteorites and earth tremors were also suggested as being the cause, and indeed would have explained some of the mystery.

But what could possibly explain the glows and beams of light seen on the mountain? They were swiftly dismissed as the villagers' imaginations, as shooting stars, or more ludicrously as people out poaching hares.⁶ Natural phenomena were also unlikely to lead to the closing of roads or large areas of mountainside by the army.

With no further information coming to light, the media soon forgot about the incident. The locals too let the matter fade from their immediate concern if not entirely from their memories. UFO researchers realized that no satisfactory explanation had been offered, and they were puzzled by reports of lights in the sky and mysterious explosions.

But in 1974 UFO crash-retrievals were barely mentioned in the UFO literature, and there was no immediate template for events in the Berwyns to fit into. Various UFO journals reported the events at the time, but no investigation was undertaken and no real conclusions were offered.

STRANGER CLAIMS

But shadowy forces appeared to be at work. Within months of the event, UFO investigators in the north of England began to receive official-looking documents from a group called Aerial Phenomena Enquiry Network (APEN). These documents claimed that an extraterrestrial craft had come down in the Berwyns; it was then retrieved for study by an APEN crash-retrieval team which had been on the scene within hours of the event. More significantly, APEN claimed there had been a key witness to the UFO crash. APEN hoped that this witness would undergo hypnotic regression.

Hypnotic regression was at that time virtually unknown in the British UFO community. In fact, other than its use in the 1961 Betty and Barney Hill abduction case, hypnosis was not common within ufology at that time.⁷

If the APEN people were hoaxers, they displayed an uncanny and detailed knowledge of ufology in general and the Berwyn Mountain incident in particular. Some researchers have speculated that APEN may have been part of a government cover-up, using UFO mythology to spread disinformation and so divert attention from secret weapons testing. APEN also issued other enigmatic communications in conjunction with other UFO events, notably the Rendlesham Forest case.

The Berwyn incident lay largely dormant throughout most of the 1970s and 1980s, little more than a footnote in the literature. But intriguing pieces of information did surface. Jenny Randles, a frequent visitor to the region in the late 1970s, stayed in the Llandrillo area for weeks at a time. She recalls hearing the locals speak about military activity on the mountains in the wake of some form of crash-like event. Though intrigued, Randles initially put it down to a possible "earthlight."

In *Places of Power*, Paul Devereux briefly relates the Berwyn Mountain incident, attributing the cause of the odd lights seen on and above the mountain to geophysical stresses. Known as "earthlights" to ufologists, these are literally lights formed by tectonic stresses within the earth. Devereux writes that a colleague, Keith Critchlow, was in the area several days after the incident and "fell in with scientists who were investigating the mountain." Their Geiger counter allegedly gave extraordinary readings in the vicinity of a Bronze Age stone circle known as Moel ty Uchaf, on the slopes of Cadair Berwyn.⁸

OTHERWORLDLY TALES

In the 1990s, amid growing interest in the UFO subject, the Berwyn Mountain incident was recalled. Randles lectured on the case at the 1995 *Fortean Times* UnConvention and mentioned the anomalous radiation count at the Moel ty Uchaf circle. Following her lecture she was approached by a science correspondent from the *Sunday Express*. He mentioned rumors of a leukemia cluster among children in the Bala area that had arisen in the years following the Berwyn Incident. At the time he connected it with possible leaks from the Trawsfynydd nuclear power station but could not prove it. In the light of later claims of UFO crashes or secret military hardware, it could be inferred that whatever had crashed had possibly been radioactive in nature and of sufficient strength to affect the human organism.⁹

By 1996 the incident had been featured in UFO books, several UFO magazines, and national newspapers. Television programs on Channel 4 and the Discovery Channel covered the case, and by 1997 it was the focus of an entire chapter in Nick Redfern's best-selling book about the government UFO cover-up, *A Covert Agenda*.¹⁰

The Berwyn incident was big news once again. From its humble beginnings it was now a "British Roswell" just waiting to burst. This surge of publicity brought forward new witnesses whose testimony added new and dramatic dimensions to the case.

In an article in *UFO Magazine*, veteran ufologist Tony Dodd recounted how his anonymous informant was part of a military unit put on stand-by several days before the date of the Berwyn incident. His unit was moved northward through North Wales until he and four others were sent to the village of Llandderfel to collect "two large, oblong boxes." They were ordered to take these to Porton Down in Wiltshire. Once at Porton Down, a U.K. government research establishment, the boxes were opened. Dodd's informant told him, "We were shocked to see two creatures which had been placed inside contamination suits. When the suits were fully opened it was obvious the creatures were clearly not of this world and when examined were found to be dead. What I saw in the boxes that day changed my whole concept of life."

Dodd's informant goes on to relate details of the creatures: "The bodies were about five to six feet tall, humanoid in shape but so thin they looked almost skeletal with covered skin."

The military man did not actually see a crashed UFO himself but claimed, "Sometime later we joined up with the other elements of our unit, who informed us that they had also transported bodies of 'alien beings' to Porton Down, but said their cargo was still alive."¹¹

A NEW INVESTIGATION

This interest by the media, together with the claims made by such researchers as Jenny Randles, Nick Redfern, Tony Dodd, and Margaret Fry, led me to reinvestigate the Berwyn Mountain incident in 1998. There was a wealth of information available, and I reasoned that somewhere, amid the accounts of the witnesses and the stories of the ufologists, lay the key to what really happened on that January night in 1974.

It is necessary to follow what has been called, since Watergate, the "paper trail." This means that any event, even a secret one, generates records. It seemed reasonable that an event of the magnitude of the Berwyn Mountain incident would have left at least some trace in official records, however small or obscure. But those ufologists who had pursued the case up to 1997 had not followed this line of inquiry, asserting that either the documentation no longer existed or it was part of the cover-up. They clearly hadn't looked hard enough, because I found a wealth of official documentation from a variety of sources. I used it, together with witness statements, to reconstruct the events of January 23, 1974. The results of that reinvestigation follow.

In *A Covert Agenda*, Redfern suggested that the numerous phantom helicopters seen in the months leading up

to the incident were flown by military UFO crash-retrieval teams. Redfern also claimed they had received advance knowledge of a UFO landing and were on permanent standby. "Perhaps," he wrote, "the idea of a joint CIA-Ministry of Defense project designed to respond on a quick reaction basis to UFO incidents should be considered."¹²

But the phantom-helicopter story is a red herring. Though a number of people had described the phenomenon as a "helicopter," a motif quickly seized upon by the media, most witnesses were in fact describing an unknown light of many shapes and colors. The "phantom helicopter" was more unidentified aerial phenomenon than unidentified flying object—a big difference. Some genuine helicopters were proved to be responsible for some sightings, but the rest remained unexplained.¹³ Additionally, the phenomenon was not seen in the Bala area, and there is no real connection between the phantom helicopters and the Berwyn Mountain incident, other than the circumstantial link Redfern makes. During my research, I discussed this in some depth with him. He stands by his published link between the phantom helicopter and the Berwyn Mountain incident. But in correspondence he qualified his belief with this statement: "All I was really trying to do was get people thinking about what might have taken place—nothing more."¹⁴

January 23, 1974, was a strange night by anyone's standards. In retrospect, it was one of those evenings when nature was staging a *son et lumière* display on a scale rarely seen. Witnesses in the villages surrounding the Berwyn Mountains reported seeing a variety of aerial phenomena that night. Besides the odd lights seen on the mountain itself, their reports and those of the media describe at least four incandescent balls of light that streaked across the Welsh skies between 7:30 and 10:00 that night. To the villagers of North Wales, they were UFOs—literally unidentified flying objects—and they described them in terms that make them sound highly unusual. One farmer, for example, described what he saw in these terms:

I saw this object coming along the mountain, about the size of a bus really, white in the middle; it came across the mountain and dipped. I thought it was going to crash.

A dramatic description that certainly echoes many UFO accounts. But there is a mundane explanation for farmer Williams's sighting and all the other aerial phenomena seen that evening. Records kept by the Astronomy Department at Leicester University, among other places, show that a number of outstanding bolide meteors were seen that night. These coincided with the approximate times given by witnesses in North Wales. The first was at 7:25, followed by another at 8:15. The third, at 8:30, coincided with the centerpiece of the evening's events. And yet another, the most dramatic of all, was seen at 9:55.¹⁵ Bolidés are considerably brighter and longer-lived than ordinary shooting stars. They can appear to be very low, depending on the position of the witness, and often trail sparks of blue and

green across the sky. Bolidés are responsible for many misperceptions of UFOs and even fool the emergency services, which are often called out to plane crashes only to discover the witnesses had seen a bright bolide meteor.

At exactly 8:38, the Bala area was rocked by a huge explosion, closely followed by a deep rumbling. One witness recalled it as being "like a lorry running into a house." Crockery rattled, furniture moved, and walls rippled slightly. Some people were certain it was a plane crash on the mountains. Others, older residents of the area, recalled earth tremors of the past and assumed it was the latest in a series of such disturbances that have taken place along the geological rift known as the Bala Fault.

This is the primary incident that has subsequently caused many UFO investigators, and the readers of their books and articles, to suggest and believe that a UFO crashed. In effect, they are saying that the noise heard and the impact felt were the UFO plowing into Cadair Berwyn. The crashed-UFO story, however, only came out years after the event. At the time, confusion reigned as to what had caused the impact.

Because of reports of lights in the sky that evening, it was initially thought that a meteorite had impacted on the Berwyns. Many people across North Wales claimed to have seen a light in the sky "trailing sparks." But this was seen at 8:30, eight minutes before the explosion, and witness descriptions indicate that it was yet another bright fireball meteor. Nonetheless, in the minds of many it has become conflated with the "explosion" to create evidence of a crash.

The explosion was heard only in the Bala area, but the tremor was felt as far away as Liverpool. By 2:00 p.m. on January 24, seismologists had determined the explosion and tremor were caused by an earthquake of 4–5 on the Richter scale.¹⁶ Its epicenter was the Bala area at a depth of eight kilometers. To cause a reading of that magnitude, a solid object (meteorite or UFO) would have weighed several hundred tons and left a massive crater. Therefore, unless a UFO crashed at the exact moment of an earth tremor, it can be safely assumed that the explosion and rumblings were the result of a purely natural process.

Following the explosion, Llandrillo district nurse Pat Evans ran out into the village street. She saw no lights, but the explosion and the accounts of other villagers convinced her that something had crashed on the mountains. It took her a while to get through to the police, because the phone lines were jammed with 999 calls, but eventually she spoke to Colwyn Bay police headquarters. When they suggested it could have been a plane crash, she bundled her two young daughters into the car and set off up the mountain, intending to offer help until the emergency services arrived.

As Evans reached the point where the B4391 mountain road levels out, she was puzzled by what appeared to be a large illuminated ball of light on the hillside. Unable to identify it, she drove on for a few minutes before returning to the same spot. The light was still there. She parked and observed it for a while. A light drizzle was falling, but the

night was otherwise clear, and Evans was able to see that the ball was “large” and formed a “perfect circle.” But it didn’t appear to be three-dimensional. In an interview she recalled, “There were no flames shooting or anything like that. It was very uniform, round in shape. . . . It was . . . flat [and] round.” As she watched, the light changed color several times from red to yellow to white. Smaller lights—“fairy lights,” she called them—could be seen nearby. It was too far away to reach on foot, and so she returned home to bed.¹⁷

Many ufologists who have written about the Berwyn Mountain incident have claimed that Evans was turned back from the mountain by soldiers and police. This is untrue. The idea arose from a misunderstanding when she was first interviewed by ufologists. Angry that she has been misrepresented, Evans stated unequivocally to me in 1998 that she saw “not a living soul” on the mountain that night. More importantly a letter from her exists, predating any interview, noting that she saw no one. This fact is significant because the misreporting of her experience has lent credence to claims that a crash-retrieval team was on the mountain shortly after the explosion.¹⁸

THE GEOLOGISTS INQUIRE

Even so, what the nurse saw on the slopes of Cadair Berwyn was still crucial to any explanation of the case. I wanted further evidence untainted by time or ufologists. For that evidence I turned to records kept by the British Geological Survey in Edinburgh. The BGS records, untouched for 24 years, revealed that within days of the explosion a team of investigators had been sent to the Bala area. This, incidentally, is almost certainly the source of rumors of “officials” who came to the area, stayed in local hotels, and questioned villagers closely about the event. That is exactly what the BGS field team did. A total of six interviewers came to the area and conducted door-to-door inquiries about the event. This is the procedure by which the BGS investigates earth tremors and earthquakes. These interviewers worked to a set questionnaire that asked such questions as “Were you at all alarmed or frightened?” and “Did you hear any creaking noises?”

These and similar questions must have seemed quite odd to the locals, especially when asked by a team of outsiders who just arrived from nowhere. More than 200 witnesses were interviewed, among them Pat Evans.

The BGS field notes were enlightening. Most ufologists have always assumed that Evans must have been on the mountain almost immediately after the explosion. They use this assumption to argue that the lights surrounding the anomalous red lights she saw must have been from a pre-alerted crash retrieval team, since no one else could have got on the mountain so quickly after the “crash.”

But the BGS records from her 1974 interview are specific about the time and say she “left house during ‘Till Death.’” Taking “Till Death” to be a reference to the

popular TV sit-com “Til Death Us Do Part,” I checked the TV schedules. Sure enough, “Til Death Us Do Part” had started at 9:30 that night. It was the only post-8:30 sit-com that evening. Knowing that Evans left the house after 9:30 means she would have observed the anomalous light sometime after 9:40, an hour later than previously thought. That hour’s difference is crucial.¹⁹

Meanwhile, 14-year-old farmer’s son Huw Thomas was also watching TV that night. At about 9:20, he answered the door to find several policemen in the farmyard. They wanted to commandeer the farm’s Land Rover, saying a plane had crashed up on the mountain. Thomas’s parents were out, so with his neighbor Enoch driving they set off up a track leading to the mountain, other police following in a car. As they neared the mountain gate they had to waste valuable time moving a car that blocked the road. Thomas recognized the car as belonging to local poachers. Once through the mountain gate, several policemen spread out on foot with flashlights, while the Land Rover and police car drove slowly up the track.²⁰

The time it took Thomas to speak to the police, load the Land Rover, drive up to the mountain, and move a car from the road would place the police search team on the lower slopes of Cadair Berwyn at about 9:40.

The BGS also interviewed one of the poachers whose car young Thomas had moved. This interview confirmed their time and position and states that the poachers “carried on work for 45 minutes [after the explosion] and were almost back at the car when met party [police, etc.] coming up.”²¹ Now a farmer in his own right, Thomas confirmed this meeting in a 1998 interview.²²

That the search party comprised of police and farmers met the poachers as they went up the mountain is further backed up by other BGS materials. Besides interviews, the BGS records also contained an Ordnance Survey map on which important witness locations and sightings of lights were plotted. This map was a revelation. It showed the anomalous light seen by the nurse, the location of the poachers, and the police search party to be all in the same small area of hillside. And as already noted, the times given to the BGS by all three parties place them there at the same time.

The logical conclusion is inescapable. Neither Huw Thomas nor the police saw the light that the nurse reported. Conversely, the nurse did see the police, though she didn’t realize it at the time. The drawing on her BGS notes clearly shows and describes “vehicles” and “torch [flash] lights.” This was the search party. Between them, very close to both, is the anomalous light source. Whatever she was seeing must have been visible to the search team and the poachers. So either the farmer and police lied about what they saw to the BGS in 1974 and to me in 1998, or it wasn’t noteworthy at the time.

But what was it? There is one possibility to consider. BGS notes also confirmed that the poachers were using powerful lamps made from spotlights powered by car

batteries. Pat Evans recalled that the weather was clear but drizzling. Lights seen in those conditions can appear to change color and size by refraction and to glow. As for the size, which she described as larger than vehicle lights, this may be a perceptual trick. Remember that Evans was looking across a dark mountainside with no visual points of reference and expecting to see a plane crash or some other scene of devastation. On the evidence available, it is certain that the nurse saw the poachers with their spotlights at the point they met and talked with the police.

Some ufologists argue that although bolides were seen throughout the evening, the beams of light seen on the mountain immediately after the explosion were not astronomical in origin and were connected to the UFO crash. Several of the BGS notes refer to people seeing these beams "on the brow" of the hill, "sometimes on and sometimes off but always vertically into sky." Another witness saw one beam "processing about the vertical." These accounts were puzzling until I looked closely at the locations of the witnesses.

All the witnesses who reported seeing these "light beams" were in the village of Llandrillo at the time. The land rises sharply to the south and to an observer in the village the "brow of the hill" is not the summit ridge of the Berwyns (actually over three miles away), but the plateau area around the 548m point: the exact area in fact where the poachers with spotlights were. The BGS records note that the poachers "continued work for half an hour to forty-five minutes" after the 8:38 earth tremor, and it was early in this time period that the beams were seen. Some villagers were convinced that poachers' lamps couldn't be responsible for the beams; others were not so sure. One witness told the BGS he had seen the poachers' lights on previous occasions, and they were exactly the same as the beams seen that night.

This theory may appear to be debunking or to be twisting the facts to fit a theory. But surely it is significant that poachers with powerful lights were in the exact area where the beams of light were seen. When questioned by the police, the poachers insisted their lights were not responsible, that they had kept them trained on the ground. Yet they also said they had not seen anything unusual. It's reasonable to suggest that as the poachers and their spotlights were in the same location as the light beams seen from Llandrillo, it was their lights people were seeing and misperceiving, perhaps because of excitement caused by the earth tremor, perhaps because of belief in a crash of some kind.

The poachers had good reason for not wishing to own up to causing bright beams of light in the sky, since it was reports of "light beams" that partially led the police to believe an aircraft had crashed. Still, there were a small number of genuinely unexplained lights seen that evening. One witness opened her curtains immediately after the tremor to see a "big bright glow in the sky over the brow of the hill." Another saw to the southeast a "glow several times

brighter than the sun." It "came and went." Maria Williams of Llandrillo saw this white glow at the same time as the poacher's lights. Other witnesses reported a firelike glow, lasting a few seconds. Some scientists have suggested this short-lived white glow was caused by the huge tectonic stresses involved in the earth tremor. But witnesses to this were few. And because it was seen at the same time as a bright meteor and the poacher's lights, it may well be yet another misperception. Indeed, one witness described the "glow" as "twinkling . . . like a streetlamp seen through heavy rain," just how a bright lamp would appear.

THE MILITARY RESPONSE

Claims by ufologists that a military presence was on the scene immediately following the 8:38 explosion and in subsequent days also bear close examination.

As we have already seen, Evans was not stopped by soldiers or police and saw no one out on the mountain roads. She set off at 7:00 for work the following day and saw nothing unusual in the village. So how did stories of a massive police and military presence arise? To understand that, we need to return again to the official records.

Following the 8:30 earth tremor, the police opened a Major Incident Log. This log shows that the police initially thought a plane had crashed and fire and ambulance services were put on stand-by. At 9:09 the police contacted RAF Valley Mountain Rescue Team (VMRT) based at Valley on Anglesey some 75 miles away.²³ A three-man team left Valley at 9:20 and arrived at Llandrillo at 12:10 a.m. The VMRT log lists the incident as "Unidentified lights and noise on hillside" and comments, "VMRT requested to investigate lights and noise on hillside. Advance party covered relevant area with negative results. Incident produced much local excitement." The fact that VMRT deemed it necessary only to send a three-man team argues strongly against the event's being of any significance. On their arrival in Llandrillo, the mountain rescue team consulted with local police who suggested they wait until morning before initiating a search.²⁴

At 7:00 a.m. on January 24, VMRT, together with local police, searched the mountains. They found nothing and abandoned the search at 2:15 p.m., possibly following official notification that the "explosion" had been caused by an earth tremor. Neither the police or VMRT logs mention any military involvement other than the VMRT itself. Huw Thomas was again out on the Berwyns that day, acting as guide for Ron Madison, a scientist who was working on the theory that a meteorite may have impacted. Madison and Thomas recall seeing no one else on the mountain other than the police and VMRT. The intense media interest, however, led to various helicopters flying over the area throughout the week, and Madison used his contacts at RAF Valley to fly a plane over the area to take a series of photographs.²⁵

But this low level of official activity doesn't account

for reports of closed and guarded roads, a military presence, or for the aircraft and twin-engined helicopters seen overhead. Looking at the paper trail, none of the police, VMRT, or BGS documents from 1974 mention this alleged military activity. In fact, the only contemporary record of a military presence comes from the article in the *Border Counties Advertiser*, the source of rumors of bodies being brought off the mountain. In looking for an explanation to this component of the story, we need to consider two crucial factors.

First, none of the witnesses were formally interviewed by ufologists until at least 20 years after the event. And second, there had been at least one other local event that contained all those elements. On February 12, 1982, an RAF Harrier jet carrying top-secret equipment crashed on Cadair Berwyn. The RAF descended on the area in force, using Gazelle and Wessex helicopters, together with Harrier and Hercules planes, in the search. The tiny village of Llandrillo was the center for this activity and was alive with RAF trucks and personnel for several days. The crash site was sealed off and guarded until the wreckage could be removed.²⁶ Additionally, there was another crash of a military plane, also carrying top-secret equipment, on the same mountain in 1971 or 1972, two years before the alleged UFO crash. Again the area was sealed off with a large military presence. It is almost certain that these incidents, at the same time of year on the same mountain, were conflated with the 1974 events.

But, proponents of the crash of a genuine alien craft say, what about the military informants who came out of the woodwork in 1996 claiming intimate knowledge of and participation in the crash retrieval? Initially this strand of the story seemed promising. As it turned out, however, these "military informants" who contacted researchers Nick Redfern, Margaret Fry, and Tony Dodd did so only after the story had been in a 1996 issue of *UFO Magazine*. They fueled the controversy surrounding the story, offering much speculation but no verifiable fact. Redfern has recently told me that his informant's telephone number has been disconnected, while Dodd refuses to expand on the identity or veracity of his contact. A close reading of Dodd's account yields more questions than answers. If the military had obtained aliens, alive or dead, would they really transport them by truck? Surely a helicopter would have been the fastest, most efficient, and most secret form of transport. Porton Down, the research establishment to which they were taken, would hardly compromise security or contamination by opening the boxes in the presence of what were essentially the delivery boys. Until these ufologists can back their claims up with substantial evidence, they remain unsubstantiated anecdotes, interesting but inconsequential to the solution of the case.

These "revelations" came also at a time when several British ufologists were being contacted by alleged military sources offering secret UFO-related information, none of which amounted to anything tangible. Researcher Kevin

McClure suggested that this was a well-organized hoax, basing his suppositions on the number of contacts made within a short timespan and the absolute absence of hard proof. APEN, the organization that circulated quasiofficial documents following the Berwyn Mountain incident, is widely regarded by most serious ufologists to have been a hoax perpetrated by ufologists on ufologists. This sort of hoax is not new to the UFO community, the most famous of the hoaxed documents being the MJ-12 papers which fooled some ufologists for more than a decade.

LOOSE ENDS

Despite the wealth of evidence to the contrary, Jenny Randles is not convinced that the Berwyn incident is completely solved. She cites the alleged anomalous radiation readings and the rumor of a leukemia cluster as possible evidence that the incident may have been a military accident, perhaps one involving a radioactive missile. Yet there are problems with Jenny's interpretation. The radiation readings taken at the Moel ty Uchaf circle in 1974 were a one-off. To have any scientific relevance at all, a series of Geiger counter readings prior and subsequent to the 1974 event would be required. As for the alleged leukemia cluster, there is no evidence to support its existence. Inquiries at the records of the National Radiological Protection Board, Greenpeace, a former radiation monitor at the Trawsfynydd Nuclear Power Station, and the archives of local papers did not reveal so much as a hint of a leukemia cluster.

In his latest book *Cosmic Crashes*, UFO researcher Nick Redfern again writes about the Berwyn Mountain incident. Redfern offers no new evidence in support of the quote cited at the beginning of this article. He further muddies the facts surrounding the case by inferring a connection with another alleged UFO crash incident that took place in Staffordshire, on Cannock Chase, a few days prior to the Berwyn Mountain event. This tenuous link clearly demonstrates how a UFO case, particularly an alleged crash-retrieval, can mutate and grow like a virus, constantly accreting new information and theories, if not actually any hard fact. There is no better subject for the study of living folklore than the genesis and development of a UFO crash-retrieval legend.²⁷

So that's where the Berwyn case stands in 1999. There are still a few loose ends and nagging uncertainties; but the symmetry of any UFO case is rarely complete, especially when it is not properly investigated for 25 years. However, I think the account I have given is the best, dare I say it, "explanation" for the disparate events that coalesced into the Berwyn Mountain UFO crash. Of course, there are those who will still choose to believe an extraterrestrial UFO crashed on Cadair Berwyn. They will continue to insist that documents have been falsified, that witnesses have been misquoted, and that a gigantic cover-up exists. That, of

(continued on page 30)

THE MYSTERY OF HOWDEN MOORS:

PART TWO

BY DAVID CLARKE AND MARTIN JEFFREY

Editor's note: Part One of this report appeared in the Summer 1999 IUR, pp. 13–20. On March 25, 1997, an English newspaper, the Sheffield Star, reported, "Emergency services from four counties were today involved in a massive operation to solve an X-Files style air crash riddle in South Yorkshire. Police treated the reports seriously because callers reporting the incident were so specific—even though air traffic authorities had no official reports of missing aircraft." Witnesses reported flashes and plumes of smoke as well as two explosive sounds like sonic booms. A British Geological Survey station at Leeds University confirmed that unexplained sonic booms had occurred in the area. Despite extensive searches, no crashed plane could be found. In the days and weeks that followed, rumors of UFOs and crashed spacecraft began to circulate, along with stories that a UFO was tracked on radar that evening. Soon, in the authors' words, the Howden Moors affair had become "one of the most controversial incidents in British ufology."

The British Geological Survey theorized that the unexplained sonic events recorded in the Sheffield area were related to a bolide meteor or manmade space junk burning up on reentry into the earth's atmosphere. Meteorites and space debris, however, were apparently ruled out as a cause of the March 24 sightings by the UK's tracking station at RAF Fylingdales, a fact confirmed by the base's public-relations officer in the summer of 1998. At no time since the search operation in March 1997 have the British authorities sought to explain the Howden Moors incident as bolide-related. In fact, Jacqueline Mitton of the Royal Astronomical Association at Cambridge University proposed it.

Meteor showers and bright bolides burning up in the atmosphere can clearly be mistaken for more exotic phenomena. Often spectacular and unexpected, they can be

accompanied by loud explosions and other optical effects. It is little wonder that sightings of these celestial phenomena often result in calls to the emergency services and searches for aircraft presumed to have been lost.

On December 1, 1997, just nine months after the Howden Moors incident, South Yorkshire Police scrambled their new force helicopter to search the Pennine Moors once again following reports of a "blazing object" crashing to earth. On this occasion John Barker, a resident of Crow Edge, a village high up on the moors at the border with West Yorkshire, called police after observing what he described as "the oddest thing I have ever seen in the sky." He said:

I was sitting in the living room of my house facing the moors when I saw what looked like a multitude of colored lights like an oxyacetylene torch through the window. It had all colors of the rainbow including orange, yellow, magenta, and green, and I could see burning debris dropping from it. It seemed to drop down and land on the moors, and then it just disappeared. I was concerned that it could have been a small aircraft and because of the bad weather they could be in trouble, so I called the police.

The police helicopter searched the area of fells near Crow Edge where Barker saw the lights descend, but neither visual observation nor heat-seeking equipment detected anything out of the ordinary. Later it was learned that police in Greater Manchester had received reports on the same night concerning a "bright green or yellow light with tails trailing behind" falling from the sky in Oldham, on the western edge of the Pennines. Air traffic control at Manchester Airport reported that no unidentified aircraft had shown up on its radar. Both groups of witnesses, it turned out, had observed a huge meteorite which had actually crashed to earth that night thousands of miles away, disappearing on the island of Greenland in the Arctic Ocean.

MILITARY DENIALS AND QUESTIONS

South Yorkshire Police and the Mountain Rescue Service contacted Royal Air Force sources as part of their routine checks as the search and rescue operation moved into full

*David Clarke, Ph.D., is a founding member of the Independent UFO Network. His books include *Phantoms of the Sky: UFOs—A Modern Myth?* (1990, with Andy Roberts). He works as a reporter for the Sheffield Star. Martin Jeffrey has been investigating UFOs and other anomalous phenomena since he was a teenager.*

swing around midnight on March 24. Both Chief Inspector Christine Burbeary, who was in charge of the police operation, and Det. Inspector Christine Wallace, who carried out further inquiries the following morning, were assured that no military aircraft operating that night could have triggered the reports and the explosive sounds.

The police and MRS were primarily in contact with staff at RAF Kinloss in Morayshire, Scotland, the designated Air Sea Rescue Coordination Center for the British coastline. Kinloss had authorized the scrambling of a Sea King helicopter from RAF Leconfield on Yorkshire's east coast to help the police and MRS search the moors for the plane. Kinloss was also responsible for contacting the BGS in Edinburgh at 9:30 on the morning of March 25 to determine if there was "any record of disturbance" recorded on its seismographs. At 11:45 Kinloss was notified that a sonic boom had been recorded. Kinloss' checks of radar tapes for the area had uncovered "nothing of significance." Soon after this, the RAF scaled down its operation and withdrew the Sea King.

Meanwhile, the RAF Press Office at Whitehall was denying that a military exercise of any kind had taken place the previous night. An MOD spokesman said at lunchtime that day that MOD was not involved in the investigation of the incident. He added that "nothing had been picked up on radar and an RAF plane was not responsible." Moreover, because the report concerned a low-flying aircraft, the inquiry was "a matter for the police." Police, Peak Park officials, and many others who live in the area are aware that the whole of the northern Peak District is a regular practice zone for the RAF. Indeed, Broomhead Estate manager Chris Thompson told us he clearly recalled seeing a large, military-style helicopter flying low and slow up the Ewden Valley the day before the events of March 24. Park ranger Brian Jones also remembers the overflight of an unidentified helicopter at dusk on March 24. Such testimony appears to suggest preparations for a military exercise were well underway hours before the Howden Moors incident began.

In short, MOD's bland denials were far from the truth. Many people in the Dronfield area of Derbyshire and parts of Sheffield witnessed low-flying military jets between 9:30 and 9:50 that night, just minutes before the reports of a low-altitude aircraft and an explosion triggered the search of the Howden Moors. Other reports of low-level military jets came from observers watching the Hale-Bopp comet from the Baildon Moor area of West Yorkshire, Wigan in Lancashire, and the M62 at Scammonden Dam earlier that evening. Among the testimony is that of an ex-RAF officer, John Brassington, who said he clearly heard a single-engined aircraft flying low above his home in Dronfield, followed minutes later by a pair of very low-flying jets—almost certainly Tornados. Brassington's testimony is supported by others, including Emma Maidenhead, who saw a formation of low-flying jets approach from the east and disappear toward the northern moors—where the search-and-rescue operation would be sparked minutes later.

In due course, when MOD was forced to admit a low-flying exercise had taken place that night, it stated that the operation had ended at precisely 9:30 p.m.—conveniently avoiding responsibility for the events over the northern Peak between 9:30 and 10:10. An acquaintance of Martin Jeffrey said he had seen six Tornados leaving RAF Coningsby in Lincolnshire at 8:45. Checks with the base, which is the home of 56 Squadron, found the flight log recorded four Tornados landing at 9:25 "following a routine exercise over the North Sea," but a spokesman said they would not have been over South Yorkshire. Similar denials came from RAF Waddington, Linton-upon-Ouse, and Leeming in North Yorkshire, home to three Tornado squadrons and Hawk training aircraft. The log there showed two squadrons flying over the North Sea at 4:30, with the others on detachment elsewhere.

At this stage we arranged a meeting with Helen Jackson, the Labor MP for Sheffield Hillsborough, whose constituency included the "crash" zone. She agreed that probing questions were in order.

Almost precisely one year after the incident, on March 23, 1998, Jackson submitted seven written questions in the House of Commons. Six were directed at Defense Minister George Robinson, the seventh at Home Secretary Jack Straw. A week later Jackson received denials that supersonic activity ("sonic events") associated with RAF or NATO aircraft had occurred that night. Under Secretary of State for Defense John Spellar noted that according to regulations, "all medium and high level supersonic flights are to be made over the sea." In a follow-up answer, Spellar said, "No low level flying is permitted over the Sheffield urban area, or any other major urban conurbation."

Pressure continued upon MOD with a series of direct questions to the RAF Press Office spokesmen, Alan Patterson and Flight Lt. Tom Rounds, during the spring and late summer of 1998. In October, Rounds provided a list of the 13 locations from which complaints of low-flying aircraft had been received on the night of March 24. Unfortunately, the times of the complaints were not recorded. The locations included, from the south coast of England: Sherborne, Dorchester, and Bridport in Dorset, Saltash in Cornwall, and Westbury in Wiltshire; Brecon and Sibenfro in Wales; Alyth and Arbroath in Tayside; Maryport in Cumbria; and in the Midlands from Rugeley (Staffordshire) and Aston in Birmingham. There were two complaints from Birstall, near Leicester. This information only added to the impression that a major RAF/NATO exercise was underway that evening, including a number of aircraft from bases across Britain.

Though unable to specify which squadrons were involved, Rounds said:

These complaints form a normal pattern. Most of the low flying is done in the Southwest, Wales and Scotland. We rarely do so, but we can fly inland between Manchester and Leeds. It's very high ground, up to

2,000 feet in places, and is a very risky place for us to be flying at low level, especially if there is low cloud, as this could lead us to penetrate built-in areas. An additional hazard for military flying is that above this area is airline traffic above Manchester and Sheffield. However, if the weather is good this would not stop us.

The weather was "good" on the evening of March 24, a clear and still frosty spring evening. The aircraft involved in the exercise came from RAF Marham in Norfolk, according to Patterson. He related that at least two Tornado GR1 Strike aircraft were involved in the exercise, which may have involved other military aircraft, including Jaguars, at various stages. Marham's GR1s were reconnaissance and bomber aircraft and would not be the type to be scrambled to pursue a UFO. All flights that evening were pre-booked training exercises, he said, which did involve low flying above the Peaks sometimes as low as 250 feet when conditions permitted. This type of low-level training is carried out regularly over areas such as the Scottish Highlands, the Lakes, and the Peak.

Making inquiries on our behalf, Marham's public-relations officer Ed Bulpitt established that six aircraft from 2 Squadron were involved in the exercise on March 24. These were Tornado GR1As, the photoreconnaissance version of the Tornado fighter. The first left Marham at 6:45 and landed at 9:10. A second left at 6:50 and returned at 9:00, while two others left at 7:40 and landed at 9:30, along with a fifth that had departed at 7:50. The final aircraft left at 7:55 and was safely back by 9:35. "They were on routine low-flying through the Peak District," he said, "and all returned safely."

We followed this with a direct challenge to the RAF Press spokesman over claims that there had been a cover-up over the March 24 incident. I also asked if he would confirm or deny that military jets had been scrambled to pursue a UFO, or whether one of the jets taking part in the exercise had inadvertently broken the sound barrier while chasing an unidentified target. The full text of Patterson's reply, in effect a summary of MOD's official stance on the Howden Moors incident, follows:

There has been no cover-up over this incident and I can confirm we did not scramble aircraft that night to intercept a UFO. All missions were regular pre-booked training flights. We have to fly low at night for training pilots for action in places like the Gulf, but we don't fly over the urban conurbations and would have avoided Sheffield. We don't break the sound barrier over land, and we don't fly below 250 feet although operational low flying is allowed in the Scottish Highlands as low as 100 feet.

Our pilots know very well they should not fly at twice the speed of sound over land and would face disciplinary action or even a police investigation if this was proved. It is not impossible that a pilot would

accelerate to supersonic speed in order to avoid taking action if a civil aircraft was detected in their flight path. We could only prove this by comparing radar traces from the night in question, but these would not show anything for the night in question as the exercise was over when the booms took place.

We responded to a request from the police to help them search for a crashed aircraft and we sent a helicopter along to help. We have not been chasing UFOs. It was not an exercise but rather a low flying sortie by two aircraft, booked long beforehand[,] and they were a long way away before this explosion was reported.

We would send fighters to intercept if we picked up anything on radar screens. It has happened before, not so much now but certainly in the past. Radar is constantly looking out and can spot incoming objects, and we have a requirement to defend the UK from attack, whether that be from Libya or from Mars. But often we find objects picked up on radar are caused by people in light aircraft wandering onto the screens and not having notified us.

We are only interested in establishing if there is a threat to UK airspace. We don't discount there may be unexplained phenomena in the sky, but we are not funded to investigate them. We would admit the fact if we had chased a UFO. There were no intercept missions that night. There is no cover-up. We don't know what caused the sightings and the sonic booms and the whole thing remains a mystery to us. We don't know what went on and there will always be people who seize on mysteries like this to make claims about cover-ups.

A CLAIM ABOUT A UFO COVER-UP

Self-styled UFO investigator Max Burns, in an Internet posting, proclaimed the Howden Moors case to be "one of the biggest UFO incidents in recent years involving a huge flying triangle, military jets, sonic booms, a bolide meteor, unmarked helicopters, glowing orange objects and what I hope, when you have studied the evidence, you will agree is a conspiracy on behalf of the civilian and military authorities to hide the facts from the public."

Burns contends that the case arose out of an incident whereby RAF jet interceptors were scrambled to pursue a slow-moving triangular UFO over Sheffield and the Peak District. As a result of the chase, one of the pursuing jets was attacked and destroyed by the UFO. The aircraft crashed to earth into one of the reservoirs northwest of Sheffield with the loss of at least one of the crew. Burns believes the two sonic booms were caused by a jet's exploding or an extra-terrestrial weapon's firing upon pursuing jets.

Passengers of a minibus returning to Sheffield spotted a man around 11 p.m., an hour after the reported air crash. The police report states that "when crossing the viaduct over the reservoir at Ladybower they were flagged down by a man who was walking towards Sheffield. The man said he

needed to get to Sheffield and wanted a lift. The minibus was full so it was declined. There were no parked vehicles anywhere in the area. The man smelled strongly of diesel. Said to be of eastern extraction, Indian/Pakistani and dressed in dark brown clothing."

A young man who was a passenger in the minibus told police about the incident the following morning, after hearing about the alleged plane crash on the moors. In his estimation the man was behaving "suspiciously." More than a year later, when a UFO investigator contacted him, the man was working for the RAF as a jet-engine engineer at RAF Cosford. He claimed that the "diesel" smell he sensed that night resembled the odor of the aviation fuel he was familiar with at air bases.

Burns decided that the stranger spotted that night was the pilot or copilot of the downed Tornado. He went on to write:

It is without doubt that the military are involved in a large cover-up regarding the attempted interception of the triangle [UFO], including conspiracy, the placement of cover-stories, debunking of witnesses[;] however[,] with this damning evidence from a member of the Royal Air Force who encountered the pilot or the copilot on the Snake Pass about an hour after the explosions occurred, stinking of aviation fuel within three miles of Howden Moors. . . . [T]he distance of three miles had been covered in one hour allowing for the time necessary to unhitch his parachute[,] then walking at 4 miles an hour would place him exactly where he was encountered by the mini-bus. I feel that this is without doubt the co-pilot of the Tornado jet, who was soaked in aviation fuel and was making his way to the nearest metropolis to alert the military.

Unfortunately for this speculation, the identity of the man encountered by the minibus on the Snake Pass that night was known to both the police and the Peak National Park ranger service. A spokesman for the latter said, "I understand it was a failed suicide attempt. It was someone who had driven out to the country and poured petrol or some other inflammable liquid over himself. But he had not gone through with the suicide. . . . It's the sort of thing that happens occasionally in a remote area like this."

A senior officer in the Derbyshire Police Force, based in Bakewell, confirmed the story. It is also worth noting that the witness on whose testimony Burns draws never once suggested that he might be the victim of an air crash.

There is not a single eyewitness report of a mid-air encounter between a jet and a UFO. Even if such an encounter had occurred, the considerable resulting wreckage would have been immediately spotted by the hundreds of volunteers who were searching the moors within hours of the incident.

Undeterred, Burns speculated that the downed Tornado had disappeared beneath the waters of one of the 13 reservoirs in the high moorland region between Sheffield

and Manchester. This was a rumor which had circulated the night of the incident by fire crews and others in the Strines area, but Yorkshire Water workers who checked a number of the reservoirs early that morning found no evidence of oil slicks or wreckage which would have been evident had a crash really occurred. The reservoirs, whose ownership is split between Yorkshire and Severn Trent Water Authorities, are the subject of regular, stringent, structural inspections and an array of specialist water-quality monitoring set by statute.

The half-hearted response of the military to the report of the "air crash" hardly fits the scenario envisioned by Burns. Experience with real air disasters, for the example the fatal mid-air collision between a GR1 fighter from RAF Cottesmore and a Cessna light aircraft above North Nottinghamshire in January 1999, demonstrates the rapid response of the military and the immediate mobilization of both ground and airborne RAF units. In the Howden Moors case it was more than one hour before the air search rescue base at RAF Kinloss authorized a single Sea King helicopter to help the civilian search-and-rescue effort. The testimony of both local people and the civilian mountain-rescue staff is that other than the Sea King crew, no other military personnel were involved in the ground search.

In addition, the complex nature of the scenario dreamed up by Burns begs the question of why it was necessary to invoke a Tornado crash at all. Initially, in the summer of 1997, he was claiming that a UFO had crashed into the moors, to be secretly retrieved by the military under the noses of the official search team. Though unprovable, such a rumor would have been certainly easier to promote than the current theory which conjures up a conspiracy on a massive scale, including the concealment of the death of at least one serviceperson from his or her family and from the hundreds of police and mountain-rescue volunteers on the ground.

But Max Burns was correct in one of his assertions: namely, that there *was* a military cover-up of the events of March 24, 1997, but it was not of a UFO chased by jets or of a UFO crash. More probably, it was a cover-up of a covert NATO exercise in preparation for the Gulf Crisis.

Burns used the Internet to distort and misrepresent the facts behind this incident, by ignoring testimony which did not fit the theory and in some cases willfully distorting evidence and testimony so that he could promote the case as "Britain's answer to Roswell." Sad to say, not a few people believed him.

CONCLUSIONS

The two-year investigation of the Howden Moors incident has taken many twists and turns in the pursuit of an explanation which can take into account all the many and varied facts which have subsequently come to light. The involvement of the military, initially denied but confirmed with reluctance in the House of Commons one year later, is of

crucial importance in the interpretation of what really happened that night. Until the Ministry of Defense decides to open its files and reveal the exact sequence of events, however, we are forced to draw our conclusions from the evidence of eyewitnesses and that of the emergency services provided by the police log. This leads to four conclusions:

1. There clearly *was* a military exercise taking place, with the Peak District at its focus, that night. One phase (now officially admitted) is timed from 7:30 to 9:00 p.m., when the RAF claims *all* its aircraft were safely grounded and accounted for. But the evidence from both witnesses in Derbyshire and South Yorkshire and the sonic booms recorded by the BGS suggest a covert part of this operation continued after the “booked” operations had been officially completed. Many more aircraft were involved in this exercise than officialdom has admitted, as is evident from the 13 low-flying complaints lodged with the RAF on March 24 from widely separated areas of the British coastline. It seems apparent that a formation of Tornado aircraft traveled across the Peak District on a southeast-to-northwest flight path between 9:45 and 10:00, coinciding with the first of two sonic booms recorded 12 minutes apart from the Sheffield area.

2. At least three groups of observers reported sightings of UFO-like phenomena during the course of the evening of March 24. These were the reports from 7:40 in Barnsley (uncorroborated), a second shortly before 10:00 by a single

observer in Dronfield, Derbyshire, and a third by a single witness in Stannington, Sheffield, who saw what she described as a cigar-shaped object. The proximity in time and space to the sightings seconds later in Bolsterstone of a low-flying aircraft suggests this latter observation was a misperception of an airplane flying at low altitude with an unfamiliar light configuration, as was suggested by a number of witnesses in the Bolsterstone area. Examined in the context of the sightings of military aircraft taking part in low-flying exercises after the fall of darkness, the “triangular UFOs” of which Burns makes much become misperceived Tornado fighter-bombers, delta-winged Jaguar aircraft, and other smaller training craft.

3. The most specific evidence connecting the Howden Moors incident with the military are the sonic events recorded by the BGS at precisely 9:52 and 10:06 on the evening of the 24th. MOD has claimed it has “no record” of RAF or NATO aircraft in relation to these booms, which were recorded in the Sheffield area by three separate seismograph stations in the case of the later event. However, the BGS explicitly states that the explosions “could only have been caused by a military aircraft reaching supersonic speed.” Breaches of the sound barrier above land are a serious offense that could result in a Military Police investigation followed by a court martial. In addition, the explosions that were recorded played a large part in the decision by South Yorkshire Police to launch a major air and land search; in American dollars, the cost to taxpayers exceeded \$150,000. Evidence from some police sources and the statement of a senior manager at Yorkshire Water suggests that the RAF did admit responsibility at a later stage in the search operation. So far, however, MOD continues to deny any connection between the sonic events and the exercise, which they maintain was completed by 9:30.

4. Coincidental to the military operation, both eyewitness testimony and the police log tell us that at least one, possibly two, light civilian aircraft were operating in the sky above north Sheffield and northeast Derbyshire, both before and after the events that triggered the search operation, coinciding with the second sonic boom. One of these fixed-wing, propeller-driven aircraft was recorded on camcorder video by witnesses in northern Sheffield shortly after 10:30, and despite a police investigation the pilot and airfield involved has never been identified. Apparently a second, somewhat larger fixed-wing aircraft followed a flight path across the north of the city shortly after 10:00, flying east to west along the Ewden Valley at an altitude below 500 feet and disappeared in a northwesterly direction over the Midhope Moors toward the Howden Reservoir. A light aircraft could not accelerate to supersonic speed; therefore, there must have been two aircraft in the same area at that time. It follows that the aircraft reported in the vicinity of the Bolsterstone that night must have been either a civilian aircraft flying illegally without having filed a flight path or a military aircraft which was playing a role in the covert military operation. ♦

PLANET IN ORBIT AROUND BINARY STAR SYSTEM

Astronomers have announced they have found evidence of the first known planet orbiting a pair of stars. Previously, planets have been found circling only single stars.

The Microlensing Planet Search (MPS) project, led by David Bennett and Sun Hong Rhie of the University of Notre Dame, used a technique called gravitational microlensing that may have revealed a planet about three times the mass of Jupiter orbiting a binary star system. The researchers, supported by the National Science Foundation (NSF), NASA, and the Research Corporation, reported their results in the November 4 issue of *Nature*.

“Between half and two-thirds of the stars in our solar neighborhood are known to be members of binary or multiple star systems,” said Morris Aizenman of NSF’s Astronomical Sciences Division. “To find evidence of a planet orbiting a pair of stars means there could be more planetary systems than we previously thought.” Astronomers have detected only about 20 planets outside our solar system, all orbiting single stars, although some of those stars are in binary systems.

The MPS astronomers analyzed data from a microlensing event that occurred in 1997, where the pattern of brightness appeared too complex to be produced by a single-star lens.

LETTERS

PERU, 1961

To the editor:

As *IUR* seems to be a focal point of serious UFO studies, I wish to relay an account of an interesting UFO sighting that I don't think has been documented. It occurred in an isolated area of the central Peruvian Andes sometime in 1961.

I was a geologist, 30 years old at the time, working for Marcona Mining Company (of San Francisco) at its iron-ore mine on the coast of Peru 250 miles south of Lima. A Peruvian friend, Manuel Montiel, of Lima, was also working at the mine, in the electrical department, I believe, and he asked me to accompany him on a weekend visit to a small but rich lead-silver mine located high in the Andes east of us (Mina Catalina Huanca). The owner, Gustavo Hernandez, of Nazca, wanted the opinion of a geologist as to the viability and potential of the mine which he had opened only two years earlier.

Reaching this mine was a real task. From Nazca, 1,800 feet above sea level, we traveled 50 miles (two hours) east over a well-graded dirt road to the Pampa de Galeras on the Altiplano at 13,500 feet elevation. This road goes on to Puquio and eventually the high Andean city of Abancay.

In the Pampa de Galeras we turned north on an unmaintained road which soon became just a track, or trail, across the gentle topography of the Altiplano at 14,000 to 15,000 feet elevation. Some of the rocks had been thrown off the track. We traveled for about 13 hours on this track, averaging five to 10 mph, about 100 miles to the mine. All people and supplies going into the mine and the ore concentrate coming out traveled this same tortuous route. My 1991 map shows an actual road at this locality now.

The mine lies on the southwest side of the wide and deep valley of the Rio Pampamarca which drains eastward into the Rio Apurimac and eventually into the Amazon. I was told the mine is located near the village of Apongo, which I located on a 1:250,000 topographic sheet at geographic coordinates 13° 50'S. lat., 73° 59'E. long., in the province of Victor Fajardo, department of Ayacucho.

In my brief stay I examined the mine and mapped the mine workings, and after my return I wrote a report, from which I draw the above details and names. A year or so later, let's say 12–18 months, which would make it sometime in 1961, a young mining engineer who had helped me map the mine visited Marcona, where I worked, and related the flying-saucer story. He said that just a few weeks (I believe) prior, a flying saucer appeared over the Catalina Huanca mine and hovered there for some time (between five and 15 minutes, maybe more). The mine is on a steep slope, and I believe from where he was, higher up, the saucer was perhaps either level with him or a maximum 45-degree angle and at only 100 meters distance.

The object was round and glowing, with windows on

the upper part. He had time to get his transit, set it up, and look at the object through the telescope. Because it was "fuzzy," he couldn't get focus on it, he said. The saucer hung around the mine area for a day or so and was seen by everyone there. (I recorded 70 persons working at the mine in my report.) The following day (as I recall) it followed an ore truck or supply truck for several hours as it exited the mine heading south over the dirt track I described. It was not seen again.

If someone is interested in following up on this 40-year-old story, I could try to help locate some of the people involved. The mine engineer, who would now be in his mid-60s, would probably be the best witness, if he's still alive. I don't remember his name, but it might be possible to track him down. I don't know what became of my friend Manuel Montiel, but he, or his family, would be easy to locate in Lima.

As to my own credibility, I have degrees from two prominent American universities, have published a number of professional articles on geology and geophysics, and have been a national officer of my professional society.

I've never seen a UFO myself, but I'm convinced such things exist, and I'd love to see one—from a distance.

S. Parker Gay Jr.
Salt Lake City, Utah

THE AIDS-ET CONNECTION

To the editor:

My compliments on the Spring 1999 issue of *IUR* devoted to abduction-related issues. I'd like to comment on secrecy in UFO research and information possibly related to the tall alien blonde in the Khoury incident (Bill Chalker, "Strange Evidence," pp. 3–16, 31).

The present situation where the APEG scientists remain anonymous exposes the negative social practice towards scientific research in ufology. This practice is dangerous because its result is (as intended) ignorance. This ignorance may extend to numerous socially important subjects, including my research on the AIDS-ET connection.

Regarding the tall alien blonde, it seems to me that the mitochondrial DNA evidence indicates her maternal ancestor shared blonde Caucasian and Mongoloid genetics, while her appearance suggests a paternal blonde Caucasian ancestor. There is a theoretical reason why aliens might employ blonde Caucasian genetics. This reason is presented in view of APEG's statements that "One might predict further that her DNA should match closely that of racial types in Finland, Iceland, or Scandinavia. . . . Hence, we may conclude with high probability, that those four human persons and the tall blonde female share a common maternal ancestor, sometime in the past 2,000–10,000

(continued on page 32)

ROSWELL—continued from page 12

Curious about the full-memory type of debris-field thin-shell material which purportedly could be easily crumpled by hand but would return to its original, unwrinkled condition when released, I asked Marcel if the pieces he held were completely free of deformation. He said they were “basically smooth but you could tell that there were some ripples in it. Not wrinkled like this one [in the photo].” His comments appear to describe a sheeting-type material having a relatively low “wrinkle factor”—say, 1 or 2. On such an admittedly arbitrary—and just made up—scale, the alleged self-restoring stuff mentioned by several eyewitnesses is a zero and the extensively crumpled laminate shown in the Fort Worth photos a 10.

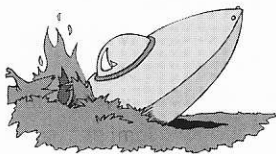
Marcel noted that the shell-like pieces he saw were not as shiny as those depicted in the photos. He also stated that the latter fragments looked larger than the ones he saw; he remembers pieces ranging from about one inch by one inch to six inches by six inches. Their edges appeared to be “torn, not cut smoothly,” similar to how the edges in the photos looked.

FINAL THOUGHTS

Marcel’s detailed recollections of the I-beam’s physical characteristics was impressive and reinforced my previous conviction that the thin-strut debris fragments were *not* balsa (or any other) wood alone or a laminated material made from aluminum (or other metal) and wood.

Having seen the “debris” photos and made my own simulated rawin-target sheeting, I am completely baffled how *anybody*—let alone highly trained personnel from the only atomic bomb-carrying base in the world—could have mistaken relatively large, highly crumpled pieces of paper-backed aluminum foil from such targets for the two different types of allegedly strange thin-shell fragments found at the Foster-ranch site. (The low-tech nature—even for 1947—of these devices is vividly apparent in Figures 5 and 6.) Roswell skeptics have still not provided a plausible answer to that nagging question.

The stiffener shown in Figure 6 appears to have a very rough surface texture. There is *no question* that this material is some kind of wood—presumably balsa wood. (Balsa wood sticks sold in hobby shops have smooth surfaces.) If Mogul Flight 4’s radar-target stiffeners were just as rough textured as their 1951–1953 counterparts, and if RAAF personnel *really* misconstrued them as something extraordinary, we would have to regard those persons as (shudder) complete incompetents. Given, however, the monumental nature of their responsibilities, that possibility is inconceivable. ♦



MOUNTAIN—continued from page 23

course, is their prerogative. But they are conclusions based on wishful thinking and not the results of an investigation.

REFERENCES

1. David Clarke and Nigel Watson, *Phantom Helicopters over Britain* (Fund for UFO Research, 1985).
2. Jenny Randles, *UFO Retrievals* (Blandford, 1995), p. 120.
3. There are numerous sources for the basic myth. Readers are directed to the following: Nick Redfern, *A Covert Agenda* (Simon & Schuster, 1997), pp. 111–124; Paul Devereux, *Earth Lights* (Turnstone Press, 1982), pp. 228–231; Jenny Randles, *UFO Reality* (Robert Hale, 1983), pp. 152–153; David Clarke, “Another One Bites the Dust,” *UFO Brigantia*, May 1989, pp. 18–20; Jenny Randles, “Britain’s Roswell,” *Sightings*, vol. 1, no. 3 (1996), pp. 12–13; Jenny Randles, “The Night the Mountain Exploded,” *IUR*, Winter 1996, pp. 9–12; Jenny Randles, “UFO Crash in North Wales,” *UFO Magazine*, September/October 1996, pp. 34–37; and Tony Dodd, *Alien Investigator* (Headline, 1999), pp. 205–210.
4. *Daily Telegraph*, January 24–25, 1974; *London Times*, January 25, 1974; *Guardian*, January 25, 1974.
5. *Border Counties Advertiser*, January 30, 1974.
6. *Wrexham Leader*, January 25, 1974.
7. Randles, *UFO Retrievals*, pp. 180–181.
8. Paul Devereux, *Places of Power* (Cassell, 1990), pp. 114–116.
9. Personal communication from Jenny Randles, April 8, 1999.
10. Redfern, *A Covert Agenda*, pp. 111–124.
11. Randles, “UFO Crash In North Wales,” pp. 34–37.
12. Redfern, *A Covert Agenda*, p. 123.
13. David Clarke and Andy Roberts, *Phantom Helicopters over Britain* (Fund for UFO Research, 1985).
14. Personal communication from Nick Redfern, October 28, 1998.
15. Letter from University of Leicester Department of Astronomy, March 25, 1974.
16. Details from British Geological Survey, Edinburgh.
17. Interview with Pat Evans, January 31, 1998.
18. Pat Evans.
19. BGS Field Diaries, ref: Mrs. Pat Evans C75, 1974.
20. Interview with Huw Thomas, January 31, 1998.
21. BGS Field Diaries, ref C64.
22. Huw Thomas.
23. Gwynedd Constabulary Major Incident Log, January 23, 1974.
24. RAF Valley Mountain Rescue Team Log, January 25, 1974.
25. Personal communication from Dr. Ron Madison, December 20, 1997.
26. M.O.D. Letter to R. Foxhall, July 14, 1998.
27. Nick Redfern, *Cosmic Crashes* (Simon & Schuster, 1999), pp. 124–144. ♦

OF INTEREST TO CUFOS ASSOCIATES

SPACELINK BOOKS (UK=1967)

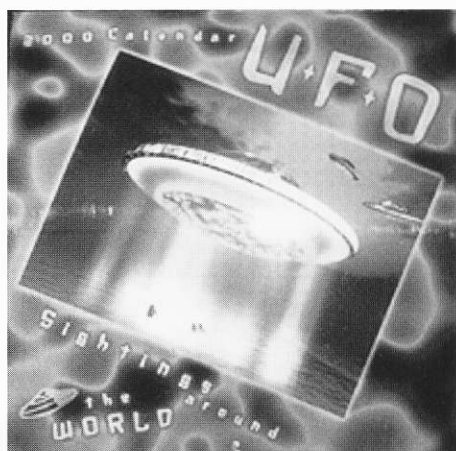
has over 400 UFO titles in stock, plus second-hand and some amazing bargains. Also: Ancient Civilizations, Conspiracies, Crop Circles, Cryptozoology, Earth Mysteries, Paranormal Magazine List. (Pay in U.S. \$\$!) FREE lists: Lionel Beer, 115 Hollybush Lane, Hampton, TW12 2QY, U.K. Tel: (+44) 20 8979 3148.

SKEPTICS UFO NEWSLETTER (SUN)

If you would like a different perspective on UFOs, perhaps you should subscribe to this bimonthly newsletter. Send \$2.00 for two sample copies to: Philip J. Klass, 404 "N" St., S.W., Washington, DC 20024.

EUROPEAN JOURNAL OF UFO AND ABDUCTION STUDIES (EJUFOAS)

Proud to announce the launch of a new peer-reviewed academic-style journal of European Ufological Research. Download subscription form at ufoinfo.com/magazines/ejufoas3.html, or write to: Editor EJUFOAS, Totton College, Water Lane, Southampton, SO40 3ZX, U.K. E-mail eufoas@totton.ac.uk.



UFO CALENDAR 2000

CUFOS still has a few calendars for the year 2000 on hand that show UFO photographs from the CUFOS files. Send \$11.95 (plus \$3.20 p/h) to: John P. Timmerman, Box 1621, Lima, OH 45802-1621.

UFO DESIGN

Tee-shirts and magnets, created by local Sedona artist inspired by sightings in the beautiful Southwest. Visit our Web site at www.etees.org, or write P.O. Box 2106, Sedona, AZ 86339. Attn: Gail Berry. Toll free: (888) 833-5673.

UFO RESEARCH STUDIES

Two unique, long-range studies: *What We Have Learned from UFO Repetitions*, and *Human Reactions to UFOs and UFOonauts Worldwide*, by veteran George D. Fawcett. Sightings, sketches, charts, statistics, references, etc. Available from Fawcett, 602 Battleground Road, Lincolnton, NC 28092. \$21 USA, \$23 overseas in U.S. dollars.

THE AGENDA: MOTIVES FOR ABDUCTIONS

The evidence is now available that proves who all aliens are and that there are people who have stopped abductions. Non-fiction. \$8.50 includes s/h. Outside USA, add \$5 for s/h. Silver Fox Press, P.O. Box 6057, Walker Branch, Roswell, NM 88202.

FLYING SAUCER DIGEST

Published since 1967, FSD is your source for no-nonsense coverage of the UFO scene. Each quarterly issue contains recent UFO reports, thoughtful speculative articles, as well as commentary and media reviews. Four issues for only \$10.00 from: FSD, 377 Race St., Berea, OH 44017.

ADVERTISE IN IUR

A maximum of 45 words is available to current Associates for \$35.00. A three-time insertion (\$105.00) earns a duplicate fourth insertion at no charge. Additional words are \$1.00 each. Send a check with ad copy to CUFOS, Box 1621, Lima, OH 45802-1621. The deadline for the Winter 1999 issue is January 10, 2000. CUFOS retains the right to refuse advertising that does not meet *IUR* standards.

LETTERS—continued from page 29

years, given known rates of substitution in mitochondrial DNA.”

The following quotes are from an abstract of a paper by Libert et al. (25 authors) published in *Human Molecular Genetics* 7, no.3 (March 1998): 399–406:

The chemokine receptor CCR5 encoded by the CMKBR5 gene located on the p21.3 region of human chromosome 3 constitutes the major co-receptor for the macrophage-tropic strains of HIV-1. The frequency of the Delta ccr5 allele was investigated in 18 European populations, and a North to South gradient was found, with the highest allele frequencies in Finnish and Mordvinian [an ethnic group found in the Middle Volga region of Russia] populations (16%) and the lowest in Sardinia (4%). . . .

It was inferred that most if not all Delta ccr5 alleles originate from a single mutation event, and that this mutation event probably took place a few thousand years ago in Northeastern Europe. It is suggested that a selection advantage is or has been associated with homo- or heterozygous carriers of the Delta ccr5 allele.

I suggest that in view of the HIV pandemics presently sweeping the earth, there is an obvious selection advantage in carrying the Delta ccr5 allele, which evidently confers resistance/immunity to HIV and which on earth is found predominantly in Finnish and Mordvinian populations, and only in Caucasian populations.

Anyone interested in knowing more about the AIDS-ET connection is invited to contact me at drpduke@juno.com, or visit homestead.com/truthsetsfree/AIDS_ET.html.

Phillip S. Duke
Omaha, Nebraska

IN DEFENSE OF HOW TO DEFEND YOURSELF

To the editor:

Thank you for publishing Jennie Zeidman's review (*IUR*, Spring 1999, p. 29) of my recent book *How to Defend Yourself Against Alien Abduction*. I appreciate Zeidman's usual objective, sharp comprehension of the subject matter, but I am puzzled why she ended the review with the statement that “especially devout Christians may benefit from [its] instruction and guidance.” I had no intention when researching the subject of resistance that any particular group in society would benefit more or less from my book. It addresses *all* groups in society, including atheists, humanists, New Agers, and persons with more traditional religious beliefs.

The book points out that the two resistance techniques that have religious overtones (two out of nine) were presented as *pragmatic* defense methods, just as the other seven are also pragmatic methods. I was not trying to teach

religion. However, perhaps she was addressing the issue that some Christians consider parapsychological and ufological subjects as demonic, and was pointing out that my book, demonstrating otherwise, opens the UFO abduction subject to readership by these particular Christian groups. If this is what she meant, I heartily agree.

If not, *IUR* readers might assume from her statement that the book benefits “devout Christians” *especially*. It does not. It can benefit everyone besieged by this phenomenon equally, and they have nine separate and distinct techniques to choose from.

Ann Druffel
Pasadena, California

ASTRONOMERS—continued from page 15

Unusual Sky Phenomenon” gives the account of Harold Hill, “an outstanding lunar observer of more than twenty years' experience in astronomical work,” who saw through his telescope near Wigan, Lancashire, on the evening of July 8, 1954, two daylight-disc UFOs:

They were separating as I watched and not only appeared to be gyrating and flashing but also exhibited a slow pendulum movement about each other in a manner similar to dancing partners!

Sometime in the 1970s, J. Allen Hynek interviewed an astronomer at New Mexico State University, who told him he had seen a group of 6–12 unusual “white spots” northwest of Clovis when he was a college student on a summer morning in 1951 or 1952. They were “loosely strung out, but not maintaining a rigid formation.” However, “the group itself had a very definite [westerly] course and speed that never wavered.” The observation lasted at least one minute, during which time one of the bright spots dropped out of the group and headed northeast until it disappeared. He told Hynek, “The weirdest thing was the fact that two of the objects seemed to have a definite relationship to each other. They seemed to be rotating around a definite center of gravity. It was very reminiscent of a binary star.”

Years later, after having forgotten the incident, his memory was jogged by seeing footage from the Tremonton, Utah, film taken by Delbert Newhouse on July 2, 1952, which shows very similar objects. Genuinely puzzled, he was willing to share his recollections with Hynek.

More recently, Chilean astronomer F. Noel shared his naked-eye observation with the readers of the *Journal of Scientific Exploration* in 1993. On the night of January 17, 1980, Noel was standing in his yard in suburban Santiago when he saw

a group of at least 30 lights distributed in a broad, symmetrical V-configuration, reminiscent of a boom-rang. . . . Since the group was in view for approximately two minutes, the mean angular velocity must have been a bit more than one degree per second.

If only more astronomers were as forthcoming! ♦